

Article

CIAFNet: An RGB-D Cross-Modal Interaction and Adaptive Fusion Network for *Camellia oleifera* Fruit Detection

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Abstract

To better address the accuracy bottleneck of RGB-only *Camellia oleifera* C. Abel fruit detection in complex orchard environments, this paper proposes CIAFNet—a dual-stream RGB-D fusion detection network—and evaluates its potential as a visual front end for relative 3D localization using sensor-measured depth and camera back projection under controlled conditions. With RGB images and depth maps as parallel dual-branch inputs, the network integrates the C3k2_PartialNetBlock for efficient intra-modal feature extraction with reduced computational redundancy, devises the cross-modal interaction and difference-aware adaptive fusion (CIDAF) module for adaptive cross-modal feature fusion, and adopts an SC-EUCB-augmented BiFPN in the neck to optimize multiscale feature aggregation and detail restoration during upsampling. Pseudo-depth maps generated from natural orchard RGB images via Depth Anything V2 were paired with RGB counterparts to build an RGB–pseudo-depth dataset. Synchronized RGB-D data collected by an Intel RealSense D435i under controlled conditions were used to quantify pseudo-to-sensor depth discrepancies and evaluate input adaptability. On the natural orchard test set, CIAFNet achieved 93.33% mAP@0.5 with only 10.49 GFLOPs and 3.80 M parameters. Second-stage fine-tuning improved CIAFNet’s adaptation to D435i-measured depth under controlled conditions. In the subsequent relative displacement consistency experiment, the mean absolute consistency errors along the X, Y, and Z axes were 3.10, 3.15, and 3.27 mm, respectively, and the mean 3D Euclidean consistency error was 5.60 mm. These results demonstrate that CIAFNet improves *Camellia oleifera* fruit detection using natural orchard RGB–pseudo-depth data and has potential as a visual front end for relative 3D localization under controlled conditions.

Keywords: *Camellia oleifera* fruit; CIAFNet; RGB-D object detection; cross-modal fusion; depth domain adaptation; 3D localization

Academic Editor: Abhilash Singh

Received: 18 July 2026

Revised: 19 August 2026

Accepted: 26 August 2026

Published: 30 August 2026

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1. Introduction

Camellia oleifera C. Abel is an economically important woody oil crop whose seeds are processed into tea oil of considerable nutritional and commercial value [1]. As plantation areas expand and seasonal labor costs rise, manual harvesting increasingly fails to meet the demands of large-scale production for efficiency, labor intensity and operational

stability. Selective harvesting robots offer a route towards mechanized and intelligent harvesting of high-value crops, for which target recognition, spatial localization, path planning and end-effector operation are tightly coupled. In natural orchards, variation in illumination, foliage occlusion, target overlap and restricted working spaces jointly compromise visual perception. Previous studies show that detection must be coupled with reliable spatial estimates to facilitate manipulator planning and end-effector grasping [2], whereas occlusion, overlapping targets, and localization errors can reduce grasping success [3]. A harvesting-oriented vision system must therefore identify fruit accurately and provide stable target regions for subsequent spatial perception.

Deep learning-based object detection is widely used for agricultural target recognition, fruit counting, phenotyping and robotic harvesting [4]. Advances in feature representation, multiscale modeling, and real-time inference have established a strong algorithmic basis for target recognition in complex agricultural scenes [5]. Real-time detectors in the YOLO family are particularly suitable for agricultural deployment because they balance speed and accuracy [6], and lightweight architectures and attention mechanisms can further improve detection under occlusion, small target, and cluttered background conditions [7]. Nevertheless, small target size, mutual occlusion, and background clutter remain major sources of false positives and missed detections in RGB-only fruit detection in natural orchards [8]. These limitations are especially pronounced for *Camellia oleifera* fruit. Severe foliage occlusion, marked scale variation, and complex background structures make it difficult for RGB appearance features alone to distinguish these fruits from foliage, background regions, and adjacent fruit [9,10]. Multiscale feature fusion can improve the representation of objects at different scales [11], and related agricultural detectors have employed feature pyramids and lightweight feature-processing modules to preserve small target details [12]. However, these methods primarily reorganize appearance features and may remain limited when RGB information alone cannot distinguish fruit from foliage, background regions, or adjacent targets [13].

RGB-D vision provides a complementary perception pathway for object detection and spatial localization. RGB images primarily encode color, texture, and semantic appearance, whereas depth maps can provide relative structural information, front-back ordering, and target boundary cues. Synchronized RGB-D data can also support spatial estimation for robotic operation [14]. The complementarity between RGB and depth may help reduce appearance ambiguity caused by partial occlusion, adjacent fruit overlap, and similarity between fruit and foliage. Noise-tolerant RGB-D fusion has consequently been investigated for outdoor fruit detection, with particular attention to the influence of unreliable depth measurements [15]. RGB-D feature fusion has also been applied to the detection of overlapping tomato clusters in complex agricultural scenes [16]. In addition, multimodal image information has been explored for recognizing *Camellia oleifera* fruits in natural environments [17]. These studies suggest that structural information can complement RGB appearance in agricultural perception, although the resulting performance depends on how heterogeneous features are represented and fused.

From the perspective of multimodal learning, simply introducing an additional modality does not necessarily yield a more discriminative representation. RGB and depth encode different types of information, and their reliability may vary across spatial locations. Early RGB-D detection research demonstrated the importance of learning modality-specific and shared feature representations rather than directly treating the two modalities as homogeneous inputs [18]. More recent RGB-D representation-learning studies have further emphasized the need to coordinate modality-specific feature extraction with cross-modal interaction [19]. In agricultural scenes, task-specific fusion architectures have been investigated to coordinate complementary appearance and structural information [20]. At a broader methodological level, token-level cross-modal exchange provides a mechanism

for selectively replacing less informative features with information from another modality [21]. Cross-modal feature rectification and fusion have also been introduced to reduce the influence of uncertain auxiliary modality responses before multimodal aggregation [22].

Existing RGB-D fusion methods generally employ feature concatenation, element-wise addition, attention-based weighting, or explicit cross-modal interaction [23]. Concatenation stacks RGB and depth features along the channel dimension and relies on subsequent convolutions to learn their joint representation, whereas element-wise addition directly superimposes spatially aligned responses. Attention-based methods reweight channels, spatial locations, or modalities, while interaction-based methods exchange or rectify information between modality-specific branches before aggregation. The limitations of these approaches arise from the spatially nonuniform reliability of RGB and depth information. Direct concatenation and addition cannot distinguish useful geometric cues from background interference [24]. Attention-based weighting improves fusion selectivity [25]; however, when the weights are estimated mainly from individual modalities or coarse global statistics, local agreement and conflict between spatially aligned RGB and depth features may remain insufficiently represented [26]. Overall, some existing RGB-D fusion strategies remain limited in explicitly modeling local agreement and discrepancy between RGB and depth features and using these relationships to guide fusion decisions. Consequently, they may have difficulty preserving complementary structural information while effectively suppressing unreliable depth responses.

Monocular depth estimation provides another means of introducing structural information when synchronized sensor depth is unavailable. Relative and metric depth estimation methods can infer spatial relationships from individual RGB images [27]. Depth Anything and Depth Anything V2 further improve the generalization of monocular depth estimation through large-scale training [28,29]. Recent surveys have summarized the development and remaining limitations of monocular depth estimation [30]. Other studies have investigated universal metric-depth estimation and monocular geometric foundation models [31,32], while diffusion-based depth prediction provides an additional estimation framework [33]. Nevertheless, DA-V2 pseudo-depth should not be regarded as sensor-measured metric depth. In this study, it is used as a relative structural prior for natural orchard RGB images that were acquired without synchronized depth measurements. Moreover, pseudo-depth and D435i-measured depth differ in representation, edge continuity, missing value patterns, and noise characteristics. Sensor-measured depth can itself contain holes and incomplete boundary information, as also discussed in depth completion research [34]. *Camellia oleifera* fruit perception therefore presents two related challenges: effective fusion of RGB and pseudo-depth during detection, and reduction of the input mismatch between pseudo-depth-based training and deployment with measured depth.

To address these challenges, this study developed CIAFNet, an RGB-D dual-stream detector integrating the C3k2_PartialNetBlock for lightweight intra-modal feature extraction, CIDAF for cross-modal interaction and adaptive fusion, and an SC-EUCB-enhanced BiFPN for multiscale feature aggregation.

The specific objectives of this study were as follows: (1) to investigate whether DA-V2 pseudo-depth can provide complementary structural information for *Camellia oleifera* fruit detection using natural orchard RGB images; (2) to assess the detection performance of CIAFNet using RGB–pseudo-depth data and its adaptation from DA-V2 pseudo-depth to D435i-measured depth under controlled conditions; and (3) to evaluate the potential of CIAFNet detections as a visual front end for relative 3D localization under controlled conditions. The overall study design and experimental workflow are summarized in Figure 1.

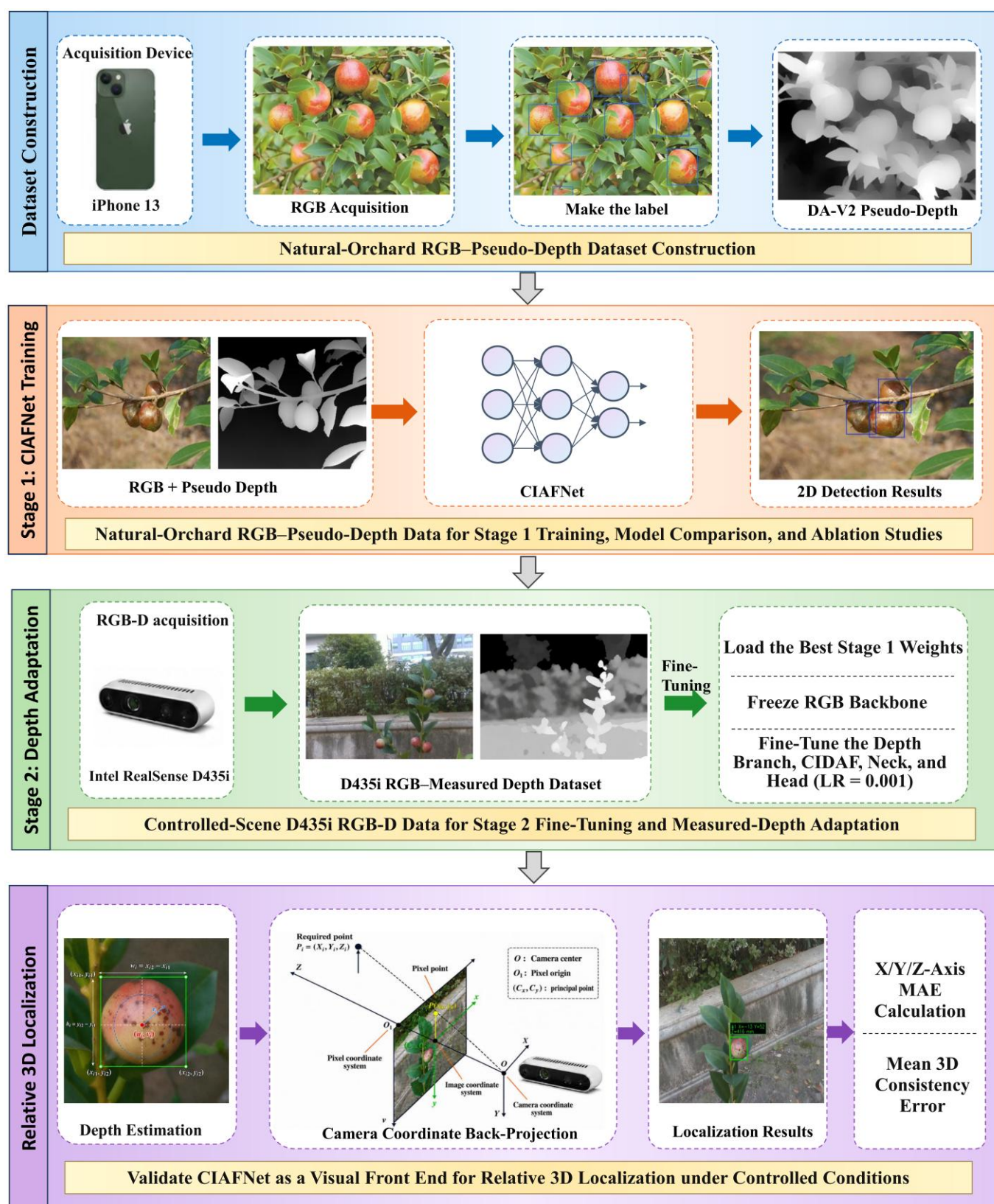


Figure 1. Graphical overview of the study design and experimental workflow.

2. Materials and Methods

2.1. Dataset Acquisition and Construction

Two complementary datasets were constructed. The natural orchard RGB–pseudo-depth dataset comprised RGB images acquired with an iPhone 13 (Apple Inc., Cupertino,

CA, USA) in real orchard environments and corresponding pseudo-depth maps generated by Depth Anything V2. It was used for CIAFNet training, model comparison, ablation studies, and pseudo-depth-assisted detection evaluation. The measured RGB-D dataset was acquired with an Intel RealSense D435i (Intel Corporation, Santa Clara, CA, USA) under controlled conditions and was used for second-stage fine-tuning and controlled domain evaluation of adaptation from DA-V2 pseudo-depth to sensor-measured depth.

Natural orchard data were collected during the 2024 *Camellia oleifera* fruit harvesting season at a plantation in Xian'an District, Xianning, Hubei Province, China. Images were acquired with an iPhone 13 in unstructured orchard environments without artificial control of illumination, background, or fruit pose. The dataset encompassed front lighting, backlighting, foliage occlusion, fruit overlap, and variable imaging distances. In total, 1758 RGB images containing 12,496 target instances were obtained. All instances were annotated as *Camellia oleifera* fruit. Bounding boxes were manually annotated using T-Rex Label (<https://trexlabel.com>; accessed on 9 April 2026) and stored in YOLO format. The dataset was divided into mutually exclusive training, validation, and test subsets at approximately 70%, 15%, and 15%, comprising 1231, 264, and 263 images, respectively.

The natural orchard dataset was originally collected as an RGB image dataset using an iPhone 13, without the synchronous deployment of a D435i or any other depth sensor during field acquisition. Consequently, no sensor-measured depth data were spatially and temporally aligned with the existing RGB frames, and such data cannot be retrospectively obtained for the same acquisition scenes. To enable these natural orchard RGB images to be used for dual-stream training, a pseudo-depth map was generated offline for each RGB image using the official Depth Anything V2 implementation with the ViT-L encoder (<https://github.com/DepthAnything/Depth-Anything-V2>; accessed on 15 April 2026). The target inference size was set to 1024 pixels, with the original aspect ratio preserved and the resized dimensions adjusted to multiples of 14. The predicted relative depth map was restored to the original RGB image resolution using bilinear interpolation. For each depth map, values outside the 1st–99th percentile range were clipped, followed by min–max normalization to map the values to [0, 1]. The normalized depth maps were stored and loaded as 8-bit single-channel grayscale images, and each pseudo-depth map was paired one-to-one with its corresponding RGB image. The resulting maps encode relative depth relationships and spatial structure rather than physical depth on an absolute metric scale. Accordingly, pseudo-depth is treated as a DA-V2-derived relative structural prior that helps distinguish structural differences among *Camellia oleifera* fruit, foliage, background regions, and adjacent fruit; it is not interpreted as a direct measurement of absolute distance. Because each pseudo-depth map is deterministically inferred from its paired RGB image, RGB and pseudo-depth should be understood as two representations of the same visual observation rather than two statistically independent sensing modalities. Therefore, while the depth branch extracts scene-related relative structural information, its learned representation may also be influenced by characteristics of the DA-V2 outputs, including their relative value distributions, spatial smoothness, and boundary continuity patterns.

An additional controlled scene RGB-D dataset was acquired with the D435i. The experimental targets consisted of artificial *Camellia oleifera* fruit models and cut *Camellia oleifera* branches rather than live plants in a natural orchard. These materials were arranged to simulate fruit targets and foliage structures under controlled conditions. The dataset contained 352 synchronized RGB-D image pairs, which were divided into approximately 60%, 20%, and 20% for training, validation, and test subsets of 211, 70, and 71 pairs, respectively. After this split, offline augmentation was applied only to the training subset and comprised random cropping, horizontal flipping, and random brightness

enhancement. Geometric transformations were applied synchronously to the RGB images, depth maps, and corresponding annotations, whereas brightness enhancement was applied only to the RGB images. Each original training pair was retained and supplemented with three augmented variants, yielding 844 image pairs for iterative training. Only the original images were retained in the validation and test subsets for model selection and independent evaluation. The D435i depth maps retained realistic sensor characteristics, including noise, holes, and edge discontinuities. This dataset does not represent RGB-D data from real *Camellia oleifera* orchards; instead, it provides measured-depth data under controlled conditions for input-mismatch analysis and adaptation verification. Natural orchard detection performance was evaluated primarily using the iPhone 13 images and their DA-V2 pseudo-depth maps. Examples of both datasets are shown in Figure 2.

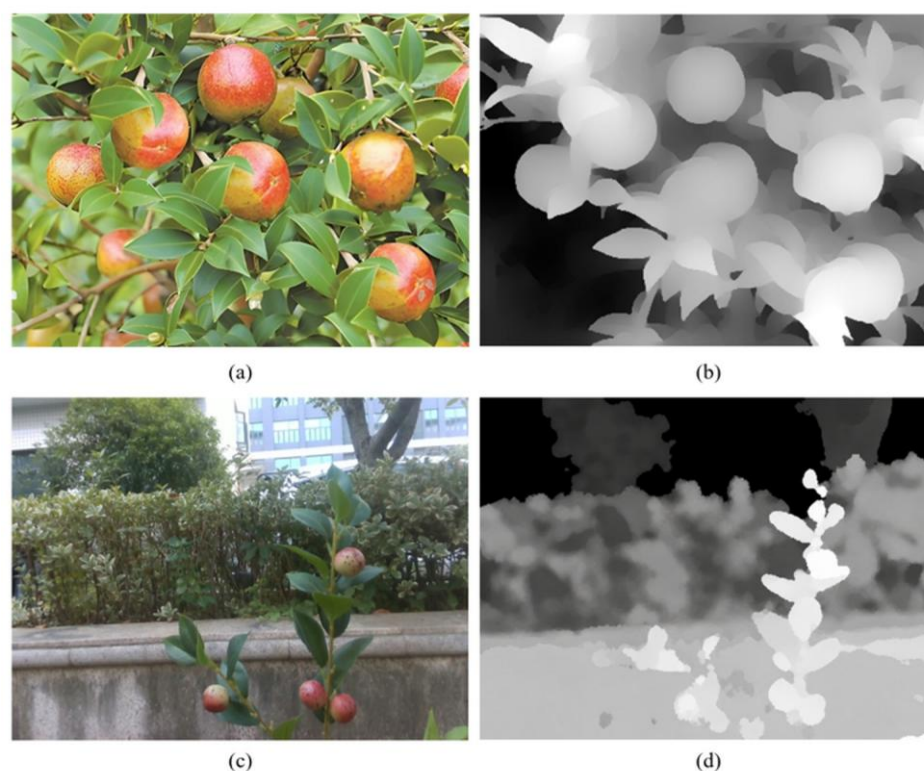


Figure 2. Examples of the two datasets used in this study: (a) an RGB image acquired in a natural orchard; (b) the corresponding pseudo-depth map generated by Depth Anything V2; (c) an RGB image acquired with the Intel RealSense D435i under controlled conditions; and (d) the corresponding sensor-measured depth map.

2.2. CIAFNet Architecture

2.2.1. Overall Architecture

CIAFNet is an RGB-D dual-stream network for *Camellia oleifera* fruit detection in natural orchards. The network comprises an RGB branch, a depth branch, a CIDAF cross-modal interaction and fusion module, an SC-EUCB-enhanced BiFPN neck, and a detection head. The RGB branch captures appearance cues, including color, texture, and edges, whereas the depth branch captures relative spatial structure among targets, foliage, background regions, and adjacent fruit. The two modalities are not simply stacked; instead, they are selectively fused by CIDAF according to their local complementarity, thereby imposing structural constraints on target representations under occlusion, fruit overlap, and background interference. The overall architecture is shown in Figure 3.

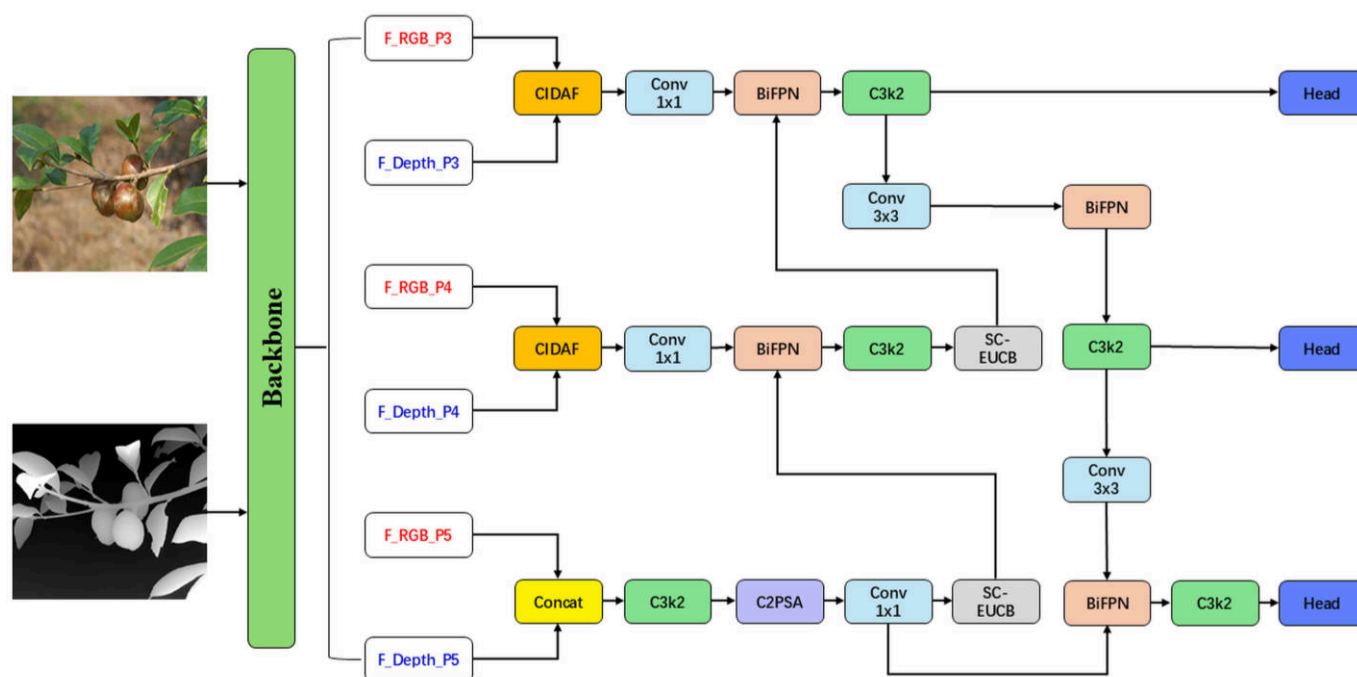


Figure 3. Architecture of the CIAFNet object-detection network.

During dual-branch feature extraction, C3k2_PartialNetBlock is inserted at the high-level stages to reduce redundant channel computation and strengthen intra-modal representations. Each branch produces features at three scales: P3, P4, and P5. P3 and P4 are passed through CIDAF to couple mid- and low-level detail with local spatial structure. P5 features are concatenated along the channel dimension and then processed by C3k2 and C2PSA to preserve high-level semantic representations. This hierarchical design treats mid- and low-level features that require fine geometric constraints differently from high-level features that require stable semantic abstraction. The dual-backbone RGB-D feature extraction architecture is shown in Figure 4.

CIDAF provides the principal mechanism for RGB-D interaction in CIAFNet. Channel alignment, discrepancy awareness, consistency modeling, and adaptive gating are used to fuse complementary information selectively and to allow depth structure to contribute directly to target representation during detection. The fused multiscale features are then supplied to a BiFPN-based neck augmented with SC-EUCB. BiFPN performs bi-directional weighted cross-scale fusion, whereas SC-EUCB improves preservation of local detail and transfer of spatial information during upsampling, thereby strengthening representations of *Camellia oleifera* fruit boundaries, small targets, and occluded regions. The detection head finally outputs class labels, confidence scores, and bounding boxes.

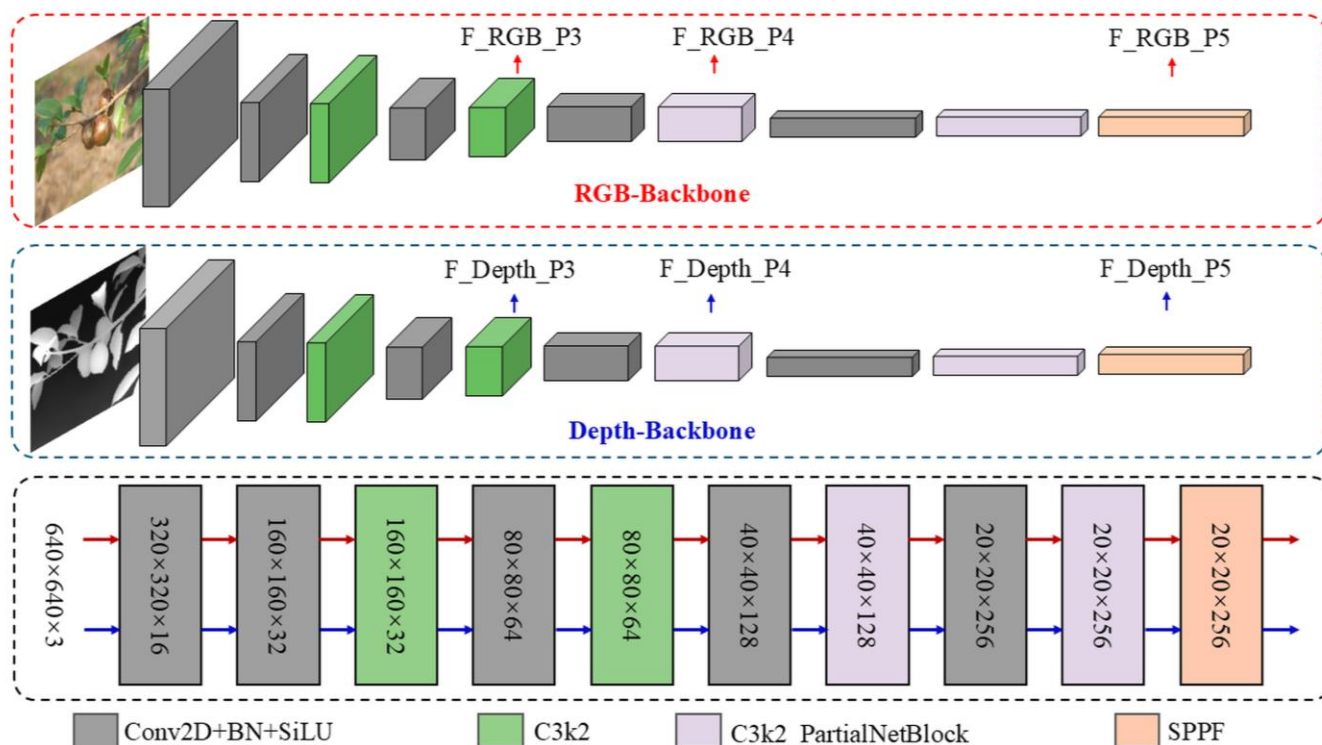


Figure 4. Schematic of the dual RGB and depth backbone feature extraction architecture.

2.2.2. RGB-D Dual-Backbone Feature Extraction

Natural orchard images of *Camellia oleifera* fruit frequently contain foliage occlusion, overlapping fruit, weakly textured targets, and complex background interference. Although dual RGB-D backbones introduce complementary structural information, they also increase computational cost. To balance representational capacity and efficiency, C3k2_PartialNetBlock is inserted into the high-level feature extraction stages of both the RGB and depth branches. The module follows the partial channel processing strategy of PartialNetBlockV1 in the Partial Channel Network [35] and is integrated with the C3k2 structure to reduce redundant spatial computation while retaining compact intra-modal representations before CIDAF fusion.

PartialNetBlockV1 exploits redundancy among feature channels by partitioning the input along the channel dimension and processing the resulting branches with different operations. For an input with C_{in} channels and a partial channel ratio r_p , approximately $C_{in} \times r_p$ channels are assigned to the spatial convolution branch, whereas the remaining $C_{in} \times (1 - r_p)$ channels retain channel context and undergo response modulation. The spatial branch applies a 3×3 convolution to extract local information, including fruit edges, texture, occlusion boundaries, and depth contours. The complementary branch retains a larger proportion of the channel context and uses global statistics-based response modulation to strengthen inter-channel representation. Unlike lightweight schemes that convolve only a subset of channels and leave the remainder unchanged, PartialNetBlockV1 preserves informative statistics in unconvolved channels. This reduces redundant dense convolution computation while maintaining a balance between computational efficiency and feature representation.

The partial channel ratio r_p controls the proportion of channels assigned to spatial convolution and therefore affects the balance between local feature extraction and computational efficiency. To examine its influence on C3k2_PartialNetBlock, r_p values of 1/8, 1/4, and 1/2 were evaluated under the same training configuration. Based on the

comparative results presented in Section 3.3, $r_p = 1/4$ was retained as the selected accuracy–efficiency trade-off among the evaluated settings rather than as the setting that maximized every detection metric.

In C3k2_PartialNetBlock, partial channel processing is embedded within the C3k2 module. Input features first pass through the standard C3k2 transformation path to model local structure and cross-channel relationships. A subset of channels then enters the spatial convolution branch to strengthen local detail, whereas the complementary branch retains and modulates channel context. The branch features are subsequently concatenated and reorganized using a 1×1 convolution to promote information exchange. This design preserves the feature extraction capacity of C3k2 while reducing redundancy through selective spatial computation. The module is illustrated in Figure 5.

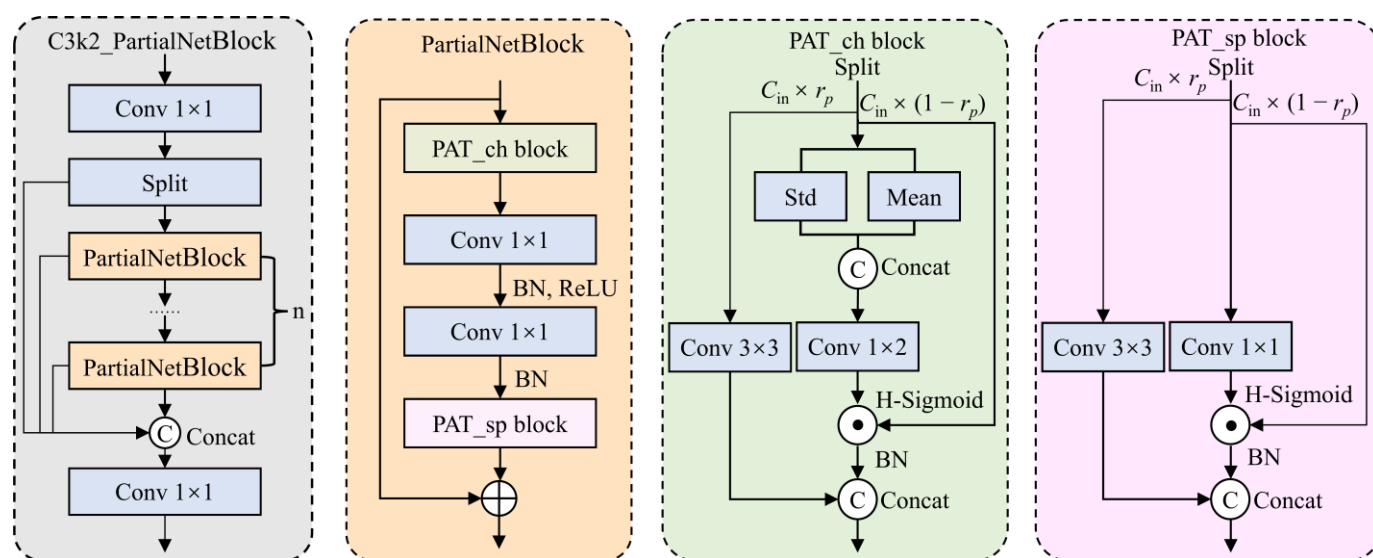


Figure 5. Architecture of C3k2_PartialNetBlock.

In CIAFNet, C3k2_PartialNetBlock is applied to the high-level feature extraction stages of both the RGB and depth backbones. In the RGB branch, it improves discrimination based on color, texture, and edge cues. In the depth branch, it emphasizes structural differences among fruit, foliage, background regions, and adjacent fruit. The resulting compact multiscale features provide more discriminative inputs for CIDAF to determine which depth information should be retained.

2.2.3. CIDAF Cross-Modal Interaction and Adaptive Fusion

CIDAF is designed to promote effective interaction and complementary information exchange between RGB and depth features. In natural orchards, RGB features may be affected by similarly colored foliage, shadows, and backlighting, whereas depth features may vary in their local reliability and discriminative value, particularly around target boundaries, occluded regions, and complex backgrounds. Conventional concatenation and element-wise addition directly aggregate the two feature streams without explicitly assessing their local relationships. Attention-based fusion can improve selectivity; however, some methods based primarily on independently estimated, global, or modality-level weights may not explicitly distinguish local cross-modal agreement from conflict. In contrast, CIDAF first aligns the RGB and depth features and explicitly constructs discrepancy and consistency representations from the paired features. These representations jointly generate spatially adaptive gates for the RGB and depth streams, after which a complementary residual branch restores information that may be weakened during gated fusion. Therefore, the principal design distinction of CIDAF lies in using explicitly

modeled local cross-modal relationships to guide adaptive fusion rather than relying only on direct aggregation or modality reweighting.

The CIDAF architecture is shown in Figure 6. The input RGB and depth features are denoted X_{RGB} and X_{Depth} and contain C_1 and C_2 channels, respectively. CIDAF first aligns the channel dimensions to obtain the RGB feature R and the depth feature D . A modal discrepancy feature F_{dis} and a modal consistency feature F_{con} , corresponding to the “Discrepancy” and “Consistency” branches in Figure 6, respectively, are then constructed as follows:

$$F_{\text{dis}} = |R - D|, F_{\text{con}} = R \odot D \quad (1)$$

Here, F_{dis} highlights locations at which the two modalities respond differently, including fruit boundaries, occlusion edges, and abrupt changes in depth structure; F_{con} reinforces candidate fruit regions supported by both modalities; and the symbol $\odot$ denotes element-wise multiplication. This discrepancy–consistency formulation treats cross-modal disagreement as local evidence to be evaluated selectively rather than amplified indiscriminately. Disagreement at a true geometric boundary can provide complementary information, whereas disagreement caused by depth noise can be suppressed by the gates. The depth branch therefore functions not merely as an additional input but as a source of structural cues that can be exploited selectively.

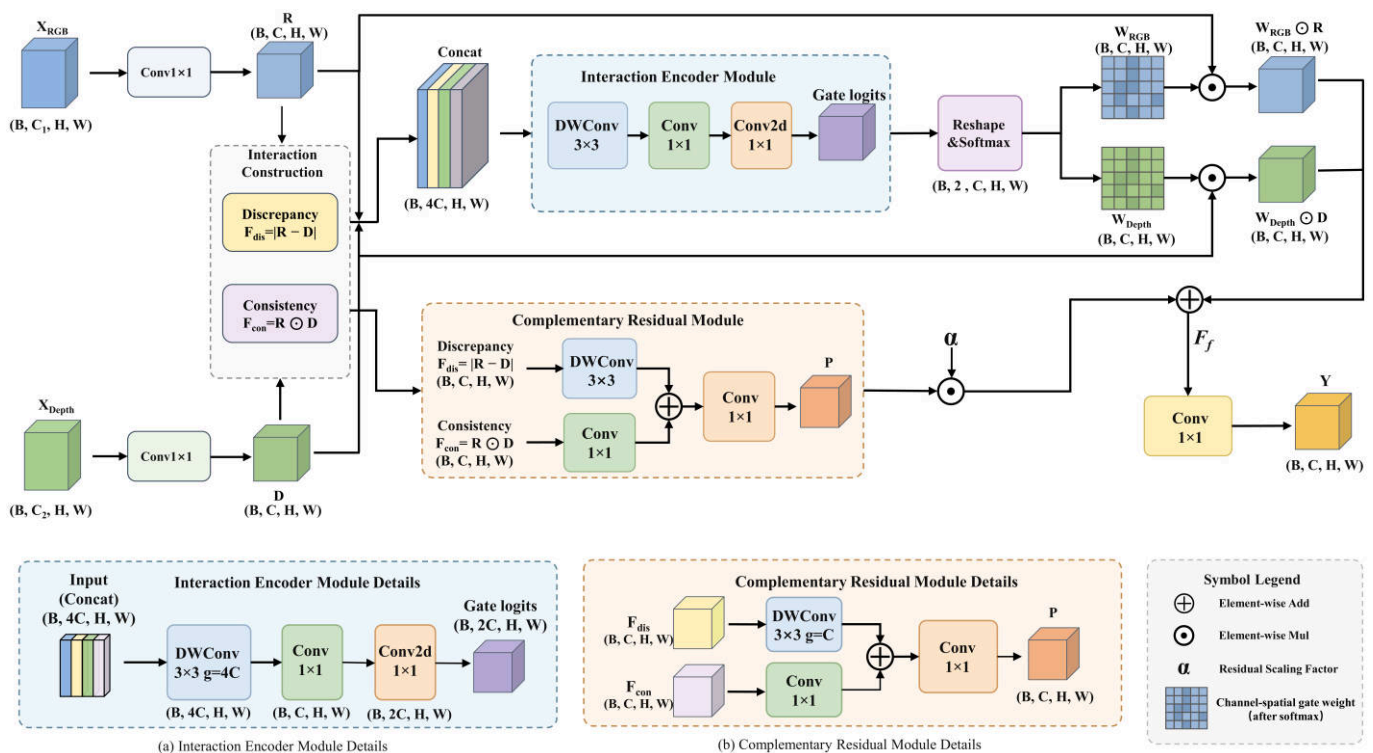


Figure 6. Architecture of the CIDAF cross-modal interaction and adaptive fusion module.

During fusion, CIDAF generates adaptive gating weights W_{RGB} and W_{Depth} from the discrepancy and consistency features. These gates adjust the contribution of each modality according to local content. When depth helps distinguish fruit, foliage, and background regions, its structural contribution is strengthened. When the depth map contains local holes, noise, or edge discontinuities, reliable structural information is retained while propagation of anomalous responses is attenuated. A complementary residual module also generates a residual feature P to restore boundary detail and local spatial structure that may be lost after gated weighting. The fused feature and the final CIDAF output are calculated as follows:

$$F_f = W_{RGB} \odot R + W_{Depth} \odot D + \alpha P, Y = \text{Conv}_{1 \times 1}(F_f) \quad (2)$$

Here, P denotes the output of the complementary residual module, and α in Equation (2) is a learnable residual scaling coefficient initialized to 0.1. At each fusion scale, $F_f, Y \in \mathbb{R}^{B \times C \times H \times W}$, where B, C, H , and W denote the batch size, number of output channels, feature map height, and feature map width, respectively. The 1×1 convolution preserves the spatial dimensions $H \times W$ and maps F_f to the CIDAF output Y with C channels. The CIDAF outputs Y at the P3 and P4 scales are subsequently fed into the corresponding input paths of the SC-EUCB-enhanced BiFPN neck. By jointly modeling RGB appearance, depth structure, and cross-modal relationships, CIDAF strengthens representations of fruit boundaries, small targets, and occluded regions while avoiding indiscriminate injection of low-confidence depth responses into the fused features.

2.2.4. Multiscale Feature-Fusion Neck

Natural orchard images can contain distant small targets, nearby overlapping fruit, and foliage-occluded targets within the same frame, producing substantial variation in target scale and local boundary quality. The neck must therefore fuse cross-scale semantic and localization information while preserving fruit boundaries, occlusion gaps, and local structure during upsampling. BiFPN is adopted for multiscale fusion, and its upsampling pathway is augmented with SC-EUCB to form an SC-EUCB-enhanced BiFPN neck. BiFPN propagates P3, P4, and P5 features through top-down and bottom-up pathways and uses learnable weights to regulate the contribution of each scale, thereby linking high-level semantics with low-level localization detail. Weighted fusion is calculated as follows:

$$F_{out} = \frac{\sum_{k=1}^K w_k F_k}{\varepsilon + \sum_{k=1}^K w_k}, w_k \geq 0 \quad (3)$$

In this expression, F_k denotes the k th input feature, w_k is the corresponding learnable fusion weight, and ε is a stabilizing term that prevents division by zero. The SC-EUCB architecture is shown in Figure 7. It extends the efficient up-convolution block (EUCB) by incorporating ShiftChannelMix and improves conventional BiFPN upsampling. The input feature is first upsampled by a factor of two to restore spatial resolution, after which a 3×3 depthwise convolution is applied for local enhancement. This operation recovers edge and texture detail that is difficult to reconstruct by interpolation alone while maintaining low computational cost. Channel shuffling and ShiftChannelMix are then applied. The channels are divided into four groups, shifted cyclically in the positive and negative directions along the height and width dimensions, and concatenated. This parameter-free operation promotes transfer of detailed responses between neighboring spatial locations and strengthens spatial information exchange among channel groups.

For *Camellia oleifera* fruit detection, SC-EUCB contributes at two levels. First, depth-wise convolution after upsampling reinforces local structure and helps recover fine-grained responses along small target boundaries and within occluded regions. Second, ShiftChannelMix expands local information exchange through lightweight shift-based mixing, thereby preserving boundary differences between densely distributed fruit and spatial gaps between fruit and foliage during cross-scale fusion. The combination of SC-EUCB and BiFPN therefore balances global scale selection with local detail recovery and supplies the detection head with more stable and discriminative multiscale features.

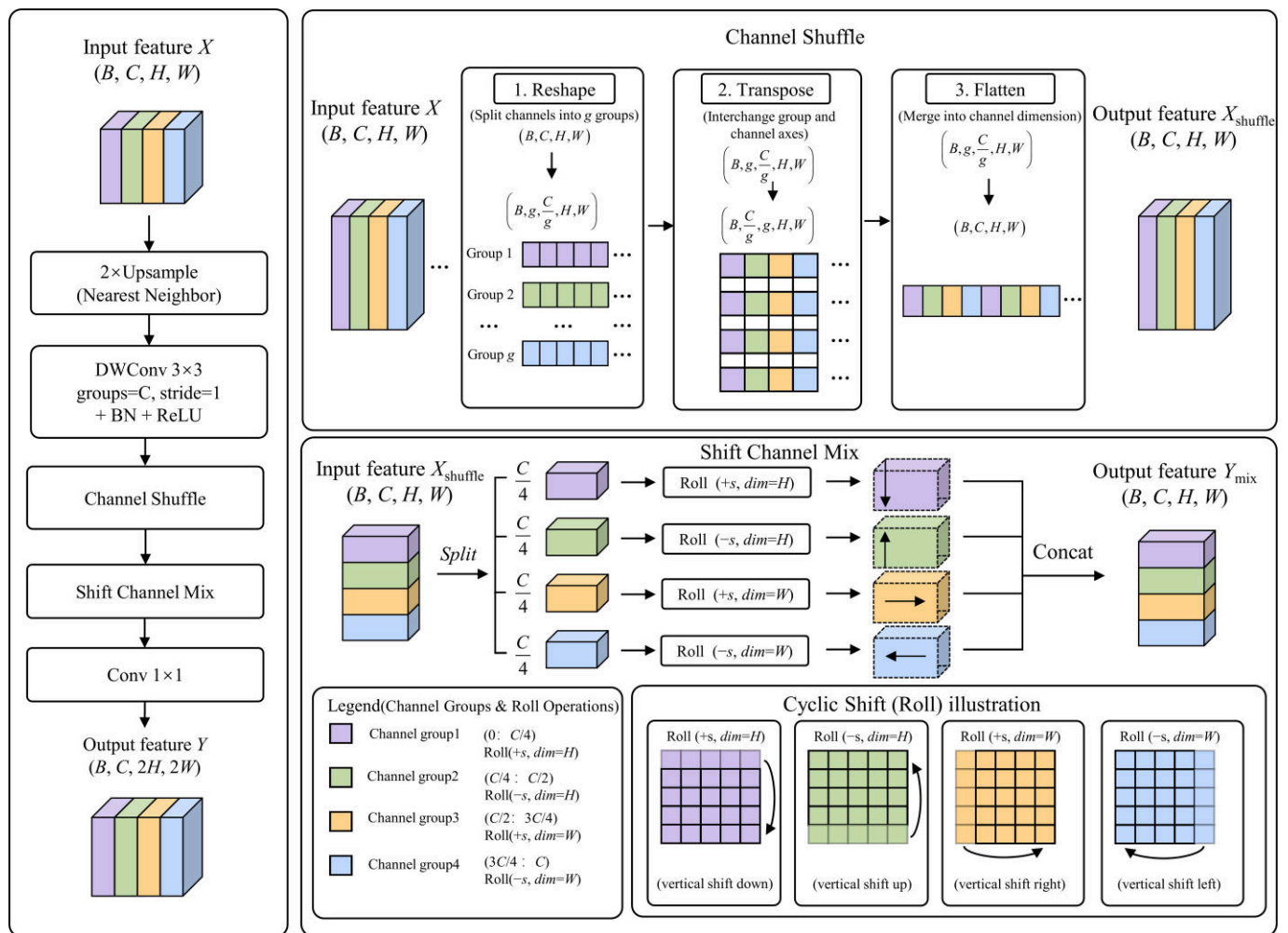


Figure 7. Architecture of the SC-EUCB module.

2.3. Circular-ROI Three-Dimensional Localization

2.3.1. Circular-ROI Coordinate Estimation

To determine whether CIAFNet detections could enable 3D localization, the coordinate-estimation pipeline used synchronized RGB and depth images acquired with the D435i, object detection, and depth back-projection (Figure 8). The D435i depth map was aligned with the RGB coordinate system, after which each RGB-D frame was supplied to CIAFNet for detection. For target i , the bounding box is denoted $B_i = (x_{i1}, y_{i1}, x_{i2}, y_{i2})$, where (x_{i1}, y_{i1}) and (x_{i2}, y_{i2}) are the upper-left and lower-right coordinates, respectively. The box width w_i , height h_i , and center pixel (u_i, v_i) were calculated, and the center was used as the pixel reference for depth sampling and back-projection. To focus the sampling region on the fruit body, a circular region of interest (ROI) centered on the bounding box center was defined using the shorter side of the box. The ROI center and radius for target i are calculated as follows:

$$u_i = \frac{(x_{i1} + x_{i2})}{2}, v_i = \frac{(y_{i1} + y_{i2})}{2}, r_i = \alpha \frac{\min(w_i, h_i)}{2} \quad (4)$$

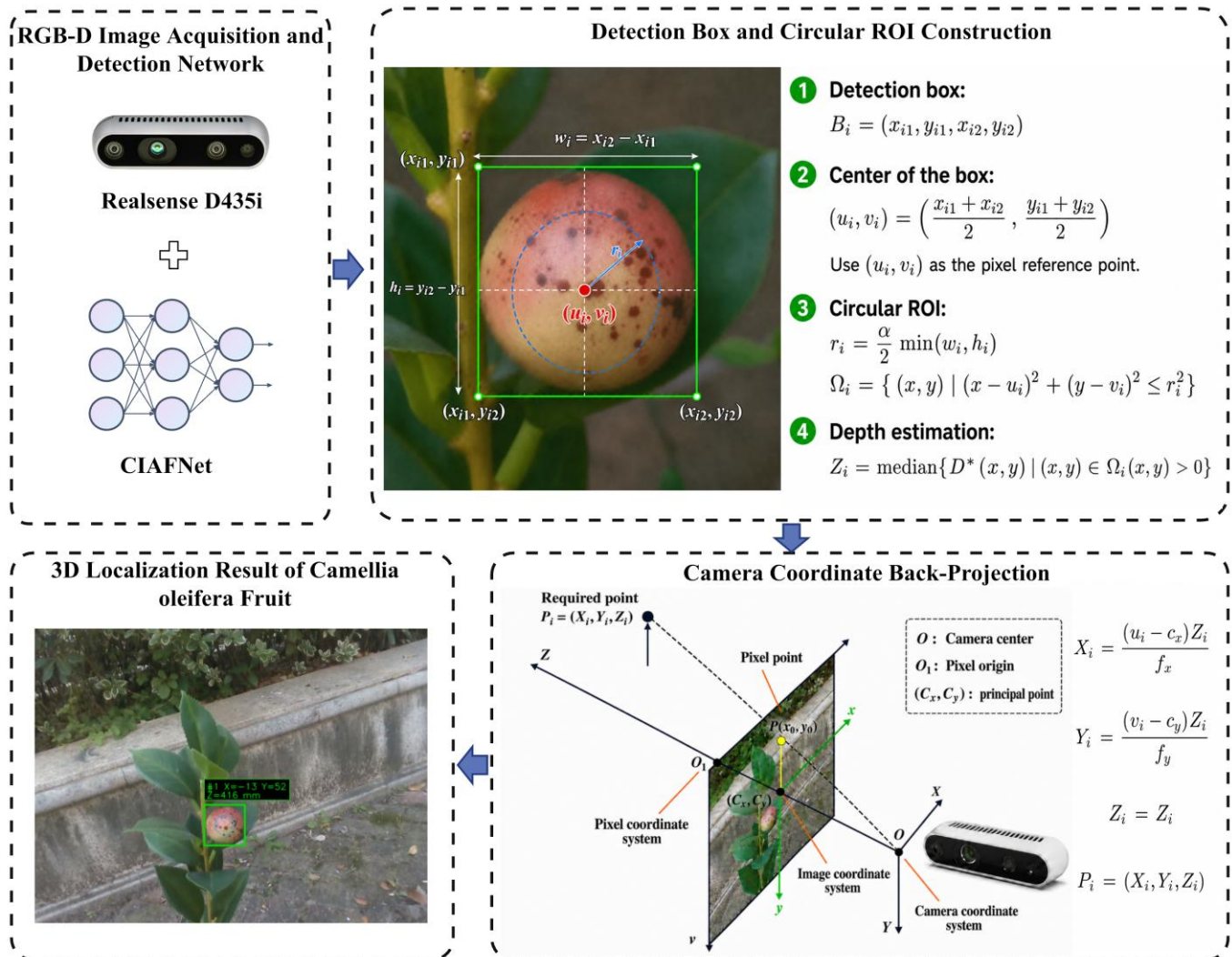


Figure 8. Circular-ROI-based 3D coordinate estimation pipeline for *Camellia oleifera* C. Abel fruit.

Here, α in Equation (4) is a scaling coefficient used to determine the radius of the circular ROI. In all experiments, α was initialized to 0.6 and heuristically adjusted within the empirical range of 0.5–0.8 according to the proportion of valid depth pixels and the stability of the depth values within the ROI. The initial value and adjustment range were selected to balance two competing sampling requirements: retaining sufficient valid depth pixels for robust median estimation and reducing contamination from background pixels and depth discontinuities near the bounding box boundary. The lower bound of 0.5 was intended to avoid an excessively small sampling region, whereas the upper bound of 0.8 was intended to limit expansion of the ROI towards the bounding box boundary. These values were task-specific empirical choices and are not claimed to be universally optimal. This adjustment retained a sufficient number of valid depth pixels while keeping the sampling region concentrated on the fruit body. For the current depth frame, the valid positive depth values within the circular ROI and the robust depth Z_i of target i , estimated using their median, are determined as follows:

$$\Omega_i = \{(x, y) \mid (x - u_i)^2 + (y - v_i)^2 \leq r_i^2\} \quad (5)$$

$$Z_i = \text{median}\{D(x, y) \mid (x, y) \in \Omega_i, D(x, y) > 0\} \quad (6)$$

Once the center pixel (u_i, v_i) and robust depth Z_i were obtained, the target was back-projected into the camera coordinate system using the camera intrinsic parameters. The

camera coordinate system follows the standard pinhole camera convention: the positive X-axis points to the right in the image plane, the positive Y-axis points downward, and the positive Z-axis points forward along the optical axis. Here, c_x and c_y denote the principal point coordinates, whereas f_x and f_y denote the focal lengths in the horizontal and vertical directions, respectively. According to the pinhole camera model, the 3D coordinates of the *Camellia oleifera* fruit target in the RGB-D camera coordinate system are calculated as follows:

$$X_i = \frac{(u_i - c_x)Z_i}{f_x}, Y_i = \frac{(v_i - c_y)Z_i}{f_y}, P_i = (X_i, Y_i, Z_i) \quad (7)$$

In this pipeline, the center of the bounding box output by CIAFNet is first mapped to a two-dimensional pixel reference. Valid depth values within the circular ROI provide the target distance in the current frame, and back-projection using the camera intrinsics yields camera coordinates $P_i = (X_i, Y_i, Z_i)$. Linking the detection output to robust depth statistics reduces coordinate fluctuations caused by background contamination within the bounding box, local occlusion, and depth discontinuities at target boundaries. The evaluation addresses the consistency of relative coordinate changes under controlled conditions rather than absolute localization accuracy in natural orchards.

2.3.2. Relative Displacement Consistency Test

A relative displacement consistency test was performed to evaluate the stability and spatial consistency of 3D coordinates derived from CIAFNet detections. Because the true absolute 3D coordinates of *Camellia oleifera* fruit relative to the camera are difficult to measure directly in natural orchards or simulated scenes, a single coordinate output cannot establish reliability. Coordinate changes were therefore compared with a known camera translation. During each experimental group, five model *Camellia oleifera* fruit targets arranged on a simulated branch remained stationary, whereas only the D435i camera mount was moved. The camera coordinates of the same target were recorded before and after movement. Because coordinates are expressed in the camera coordinate system, the expected change in target coordinates is equal in magnitude and opposite in direction to the prescribed camera translation. The difference between the observed and expected target coordinate changes was defined as the relative displacement consistency error, which quantifies the stability and spatial consistency of the visual front end rather than absolute positional accuracy.

The test was performed as follows. The *Camellia oleifera* fruit model or branch was first fixed. The D435i pose was then adjusted until the target was within the field of view and could be detected and localized stably in 3D by CIAFNet. The 3D coordinates of each target before camera movement were recorded as the initial coordinates. While the target remained stationary, the camera mount was translated along a predefined combined direction, with prescribed displacement components of 100 mm along the horizontal, vertical, and depth axes. A second RGB-D image was then acquired, and the coordinates of the same target were recorded. The observed coordinate changes along each axis were compared with the corresponding expected target coordinate changes induced by the camera translation to obtain the relative displacement consistency error. Ten experimental groups were conducted under controlled conditions. In each group, the coordinates of five model fruit targets were recorded before and after the prescribed camera translation, yielding 50 target-level relative displacement measurements in total. Group-wise errors and pooled descriptive results across all target observations were calculated. This experiment evaluated the consistency of coordinate changes under prescribed camera translation rather than the absolute positioning accuracy of individual targets.

2.4. Experimental Environment and Training Settings

All models were trained and tested in the same hardware and software environment to ensure comparability and reproducibility. An identical base configuration, including input resolution, number of training epochs, batch size, optimizer, and initial learning rate, was used for model comparisons, ablation experiments, and analyses of RGB-D modality effectiveness. The experimental environment and training settings are listed in Table 1.

A two-stage training procedure was used to separate the learning of natural orchard appearance and DA-V2-derived relative structural priors from adaptation to D435i-measured depth. Throughout this study, CIAFNet denotes the unchanged network architecture. In the first stage, CIAFNet was trained using natural orchard RGB images and their DA-V2 pseudo-depth maps to establish the RGB–pseudo-depth detection representation; the resulting model state is referred to as CIAFNet (Stage 1). In the second stage, the best CIAFNet (Stage 1) weights were loaded and fine-tuned using RGB-D data acquired with the D435i under controlled conditions; the resulting measured depth-adapted model state is referred to as CIAFNet (Stage 2). Thus, CIAFNet (Stage 1) and CIAFNet (Stage 2) represent two sequential training states of the same architecture rather than two different network structures. This sequential configuration was adopted according to the different data sources and depth input domains and is not claimed to be a globally optimal training schedule. In the second stage, the initial learning rate was reduced from 0.01 to 0.001 to allow conservative parameter updates while retaining the representations learned during the first stage. The maximum number of training epochs was kept at 150, consistent with Stage 1, to maintain a comparable training budget across the two stages. Only the RGB backbone was frozen during the second stage; the depth branch was not frozen. Freezing the RGB backbone was intended to preserve the appearance representations learned from the larger natural orchard dataset and to reduce the risk of overfitting to the smaller controlled scene D435i dataset or excessively modifying the previously learned appearance representation. The depth branch, CIDAF fusion module, SC-EUCB-enhanced BiFPN neck, and detection head remained trainable so that the depth-related representations and fusion weights could adapt to the statistical characteristics of D435i-measured depth, including sensor noise, missing values, and edge discontinuities. The remaining training settings were kept unchanged. Only the RGB images and their synchronized D435i-measured depth maps were used in the second stage; DA-V2 pseudo-depth and measured depth were not introduced simultaneously as additional channels.

The network-only inference efficiency of CIAFNet was evaluated on the workstation listed in Table 1. Before inference, the convolution and batch normalization layers were fused. All measurements were conducted using FP16 precision with a batch size of 1. Each input consisted of paired RGB and depth images resized to 640×640 pixels, forming a six-channel RGB–depth input. To minimize initialization and GPU warm-up effects, 100 preliminary forward passes were performed before timing. The latency was then measured over 1000 forward passes, with CUDA synchronization applied during timing to ensure accurate GPU latency measurement.

Table 1. Experimental environment and training parameters.

Item	Specification
Operating system	Microsoft Windows 11 Pro 64-bit
CPU	Intel Core i5-13600KF (Intel Corporation, Santa Clara, CA, USA)
GPU	NVIDIA GeForce RTX 4070 SUPER 12 GB (NVIDIA Corporation, Santa Clara, CA, USA)
Deep learning framework	PyTorch 2.5.1 + CUDA 12.1

Input resolution	640 × 640
Epochs	150
Batch size	16
Optimizer	SGD
Initial learning rate (first stage)	0.01
Initial learning rate (second stage)	0.001

3. Results and Analysis

3.1. Evaluation Metrics

CIAFNet was evaluated at three levels: detection performance, separability of fused features, and relative displacement consistency. The first two assessed the reliability of detection and the discriminability of fused representations, respectively, whereas the third determined whether the detections provided stable inputs for relative 3D coordinate output under controlled conditions.

Detection performance for *Camellia oleifera* fruit was quantified using precision (P), recall (R), mean average precision at an intersection-over-union threshold of 0.5 (mAP@0.5), and mAP averaged over thresholds from 0.5 to 0.95 (mAP@0.5:0.95). Model complexity was quantified using the number of parameters (Params) and the number of floating point operations in billions (GFLOPs). The specific calculation formulas for these evaluation metrics are as follows:

$$P = \frac{TP}{TP + FP} \quad (8)$$

$$R = \frac{TP}{TP + FN} \quad (9)$$

$$mAP@50 = \frac{1}{n} \sum_{i=1}^n AP_i \quad (10)$$

$$Params = (K_h \times K_w \times C_{in} + 1) \times C_{out} \quad (11)$$

$$GFLOPs = \frac{2 \times H_{out} \times W_{out} \times K_h \times K_w \times C_{in} \times C_{out}}{10^9} \quad (12)$$

Here, TP is the number of correctly detected *Camellia oleifera* fruit targets, FP is the number of false detections, FN is the number of missed detections, and AP_i is the average precision for class i . K_h and K_w are the convolution kernel height and width, respectively; C_{in} and C_{out} are the numbers of input and output channels; and H_{out} and W_{out} are the height and width of the output feature map.

To quantify the statistical uncertainty of the detection results, an image-level paired nonparametric bootstrap analysis was performed on the natural orchard test set containing 263 images. The trained model weights were fixed, and no model retraining was performed. In each bootstrap iteration, 263 images were sampled with replacement, and the same resampled image indices were applied to RGB-only, Baseline, and CIAFNet to preserve the paired comparison structure. The mAP@0.5 and mAP@0.5:0.95 metrics were recalculated for each resampled test set. For bootstrap iteration b , the paired difference was calculated as:

$$\Delta_b = M_{CIAFNet,b} - M_{reference,b} \quad (13)$$

where M represents mAP@0.5 or mAP@0.5:0.95, and the reference model represents RGB-only or Baseline. This procedure was repeated 5000 times. The bootstrap mean and standard deviation (SD) were calculated for each model. The mean paired difference and its

two-sided percentile-based 95% confidence interval (CI) were calculated from the paired difference distribution, with the CI defined by its 2.5th and 97.5th percentiles. A difference was considered statistically significant at the two-sided $\alpha = 0.05$ level when its 95% CI did not include zero. The bootstrap SD represents test set sampling uncertainty conditional on the fixed trained model checkpoints rather than variability across repeated training runs.

Feature separability was quantified using the silhouette coefficient (Sil) and the Davies–Bouldin index (DBI). A larger Sil indicates tighter within-class clustering and greater between-class separation, whereas a smaller DBI indicates better separation among feature clusters. The corresponding calculation formulas are as follows:

$$Sil = \frac{1}{n} \sum_{i=1}^n \frac{b_i - a_i}{\max(a_i, b_i)} \quad (14)$$

$$DBI = \frac{1}{K} \sum_{i=1}^K \max_{j \neq i} \left(\frac{S_i + S_j}{M_{ij}} \right) \quad (15)$$

Here, n is the number of samples; a_i is the mean distance from sample i to samples of the same class; b_i is the mean distance from sample i to samples in the nearest different cluster; K is the number of classes; S_i and S_j are the mean within-cluster distances; and M_{ij} is the distance between the centers of clusters i and j .

Consistency of the 3D coordinate output was quantified using the axis-wise absolute errors, the mean axis-wise absolute errors, and the mean 3D Euclidean error. The relative displacement consistency error measures the agreement between the observed change in the 3D coordinates of the same *Camellia oleifera* fruit target before and after a known camera translation and the expected target coordinate change induced by that translation. For target i , the errors along the three axes are denoted $e_{x,i}$, $e_{y,i}$, and $e_{z,i}$. Averaging over all targets yields the mean absolute errors E_x , E_y , and E_z along the X, Y, and Z axes, respectively. The mean 3D Euclidean error E_{3D} is calculated as the average Euclidean norm of the per-target error vectors, as follows:

$$e_{x,i} = |(X_i^1 - X_i^0) - \Delta X|, e_{y,i} = |(Y_i^1 - Y_i^0) - \Delta Y|, e_{z,i} = |(Z_i^1 - Z_i^0) - \Delta Z| \quad (16)$$

$$E_x = \frac{1}{N} \sum_{i=1}^N e_{x,i}, E_y = \frac{1}{N} \sum_{i=1}^N e_{y,i}, E_z = \frac{1}{N} \sum_{i=1}^N e_{z,i} \quad (17)$$

$$E_{3D} = \frac{1}{N} \sum_{i=1}^N \sqrt{e_{x,i}^2 + e_{y,i}^2 + e_{z,i}^2} \quad (18)$$

Here, (X_i^0, Y_i^0, Z_i^0) and (X_i^1, Y_i^1, Z_i^1) are the 3D coordinates of *Camellia oleifera* fruit target i before and after camera movement, respectively; ΔX , ΔY , and ΔZ denote the expected coordinate changes of the stationary target in the camera coordinate system induced by the prescribed camera translation; and N is the number of targets included in the analysis. Under the coordinate convention used here, the expected target coordinate change was (−100, +100, −100) mm. Smaller errors indicate greater consistency between the model-derived 3D coordinates and the expected coordinate change.

3.2. Comparison with Other Detection Models

Eight representative detectors were selected for comparison with CIAFNet on *Camellia oleifera* fruit detection in complex natural orchard backgrounds. YOLOv5n, YOLOv8n, YOLO11n, YOLOv13-N, YOLO26n, and RT-DETR-R18 were included as RGB-only

baselines, whereas YOLO-RGBDtea [36] and DEYOLOn [37] were included as multimodal baselines. For the modality-matched comparison, YOLO-RGBDtea, DEYOLOn, and CIAFNet (Stage 1) received the same RGB images and corresponding DA-V2 pseudo-depth maps. All models were evaluated using the same data split, input resolution, training settings, and evaluation metrics. The quantitative results are summarized in Table 2, and representative detection results are presented in Figures 9–11.

The RGB-only baselines provide conventional detection references, whereas the comparison among YOLO-RGBDtea, DEYOLOn, and CIAFNet (Stage 1) provides a modality-matched assessment under the same RGB–pseudo-depth input. CIAFNet (Stage 1) achieved a precision of 94.54%, a recall of 84.92%, an mAP@0.5 of 93.33%, and an mAP@0.5:0.95 of 73.67% (Table 2). Among the RGB–pseudo-depth models, CIAFNet (Stage 1) achieved the highest values for all four metrics. Compared with YOLO-RGBDtea, CIAFNet (Stage 1) improved precision, recall, mAP@0.5, and mAP@0.5:0.95 by 0.26, 0.29, 0.47, and 2.78 percentage points, respectively. Compared with DEYOLOn, CIAFNet (Stage 1) improved precision, recall, mAP@0.5, and mAP@0.5:0.95 by 4.55, 2.02, 3.28, and 7.67 percentage points, respectively. In this experiment, the three multimodal models shared identical inputs, using the same RGB images and pseudo-depth maps generated by DA-V2. The results under this matched input condition indicate that the performance advantage of CIAFNet cannot be attributed solely to the introduction of pseudo-depth information, which indirectly lends support to the merits of its model design.

Table 2. Comparison of *Camellia oleifera* C.Abel fruit detection performance across models.

Model	Input Modality	P/%	R/%	mAP@0.5/%	mAP@0.5:0.95/%
YOLOv5n	RGB	87.79	78.03	88.13	64.34
YOLOv8n	RGB	91.49	80.21	90.29	64.99
YOLO11n	RGB	92.05	82.33	91.11	68.32
YOLOv13-N	RGB	88.59	79.39	89.72	66.43
YOLO26n	RGB	92.31	81.19	90.64	67.84
RT-DETR-R18	RGB	89.42	80.67	89.44	65.49
YOLO-RGBDtea	RGB + DA-V2 Pseudo Depth	94.28	84.63	92.86	70.89
DEYOLOn	RGB + DA-V2 Pseudo Depth	89.99	82.90	90.05	66.00
CIAFNet	RGB + DA-V2 Pseudo Depth	94.54	84.92	93.33	73.67

Note. In the table, CIAFNet denotes the model state obtained after first-stage training on the natural orchard RGB–pseudo-depth dataset.

Figures 9–11 show model performance under close-range mild occlusion, distant complex backgrounds, and backlighting with dense foliage occlusion. Blue boxes denote model-predicted bounding boxes, red boxes denote missed detections, and red circles denote false detections. In the close-range scene, CIAFNet preserved more complete fruit boundaries. In the distant scene, fewer small fruit targets were missed. Under backlighting, shadows, and dense foliage occlusion, several *Camellia oleifera* fruit targets in shadowed regions were still detected, with fewer false detections than were produced by the comparison models. Collectively, these findings indicate that CIAFNet improves occluded target recall, small target recognition, and boundary localization stability. The visualizations provide qualitative evidence consistent with the proposed mechanism but do not independently establish the effectiveness of cross-modal fusion.

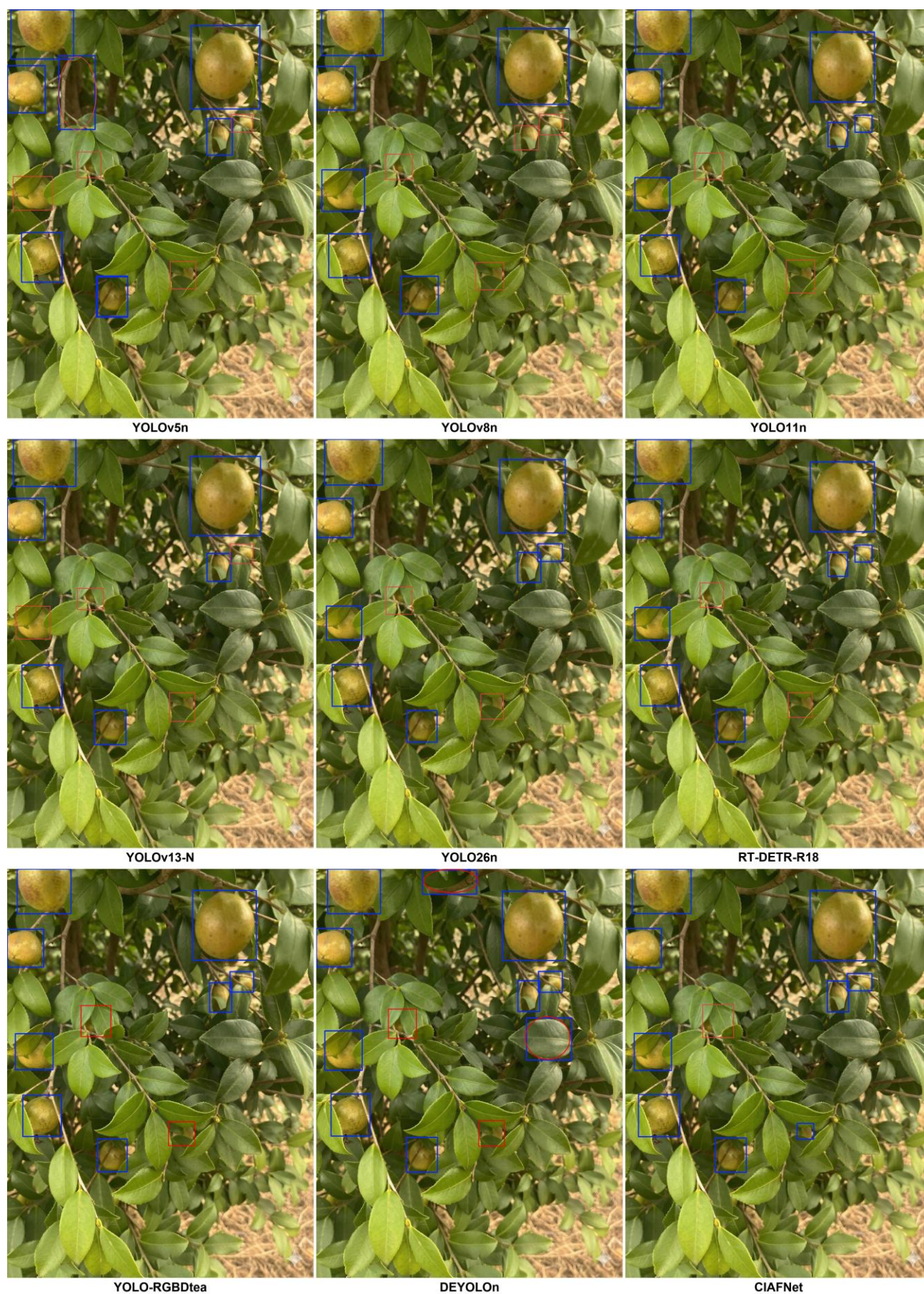


Figure 9. Comparison of detection results in a close-range scene with mild occlusion.



Figure 10. Comparison of detection results in a distant scene with a complex background.

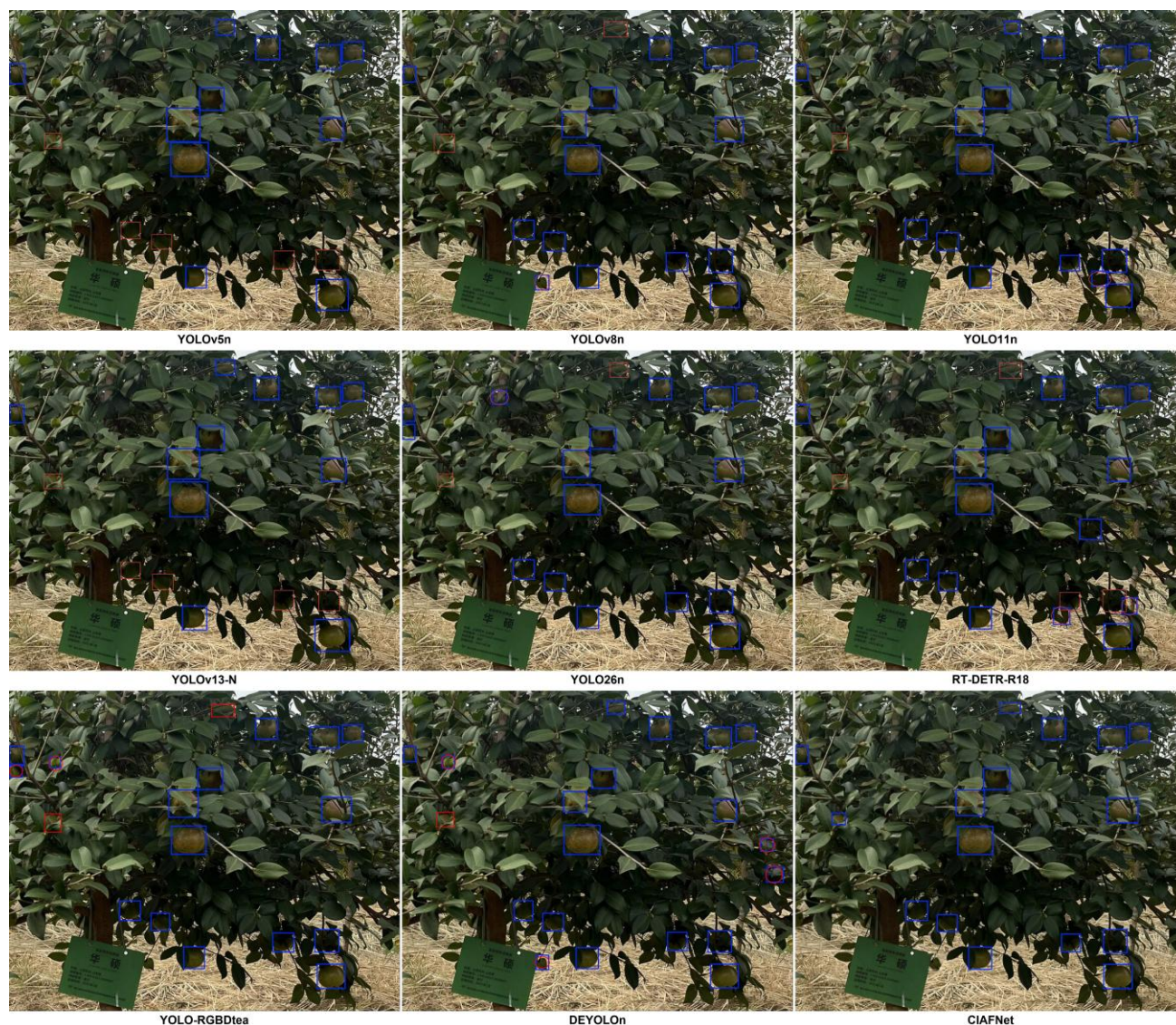


Figure 11. Comparison of detection results under backlighting, shadows, and dense foliage occlusion.

3.3. Ablation Studies of CIAFNet

Progressive ablation experiments were performed to quantify the contribution of each design component. The baseline was the basic dual-stream detector without the proposed modules. Model 1 added C3k2_PartialNetBlock to Baseline; Model 2 further added CIDAF; Model 3 additionally incorporated BiFPN for multiscale feature fusion; and the complete CIAFNet augmented BiFPN upsampling with SC-EUCB. In addition, $r_p = 1/8$, $1/4$, and $1/2$ were evaluated in Model 1 to examine the influence of the partial channel ratio on detection performance and computational efficiency. Table 3 presents the r_p sensitivity analysis, whereas Table 4 presents the progressive model ablation results. Based on the accuracy–efficiency comparison in Table 3, $r_p = 1/4$ was used in the subsequent model configurations.

Table 3. Sensitivity analysis of the channel-partition ratio in Model 1.

Model	r_p	P/%	R/%	mAP@0.5/%	mAP@0.5:0.95/%	GFLOPs
Baseline	-	92.01	84.09	91.90	71.25	12.20
Model 1	1/8	91.42	83.62	91.83	70.53	10.94
Model 1	1/4 [†]	91.78	84.05	92.15	70.66	11.07

Model 1	1/2	91.88	84.12	92.23	70.78	11.55
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Note. † indicates the setting retained in the subsequent model configurations.

Table 4. Progressive ablation results for CIAFNet.

Model	Added Module(s)	P/%	R/%	mAP@0.5/%	mAP@0.5:0.95/%	GFLOPs
Baseline	-	92.01	84.09	91.90	71.25	12.20
Model 1	+ C3k2_PartialNetBlock ($r_p = 1/4$)	91.78	84.05	92.15	70.66	11.07
Model 2	+ CIDAF	93.88	84.82	92.88	72.65	11.93
Model 3	+ BiFPN	93.92	84.74	93.05	73.11	10.35
CIAFNet	+ SC-EUCB	94.54	84.92	93.33	73.67	10.49

Note. In the table, CIAFNet denotes the model state obtained after first-stage training on the natural orchard RGB–pseudo-depth dataset.

The sensitivity analysis in Table 3 shows that increasing r_p from 1/8 to 1/2 progressively improved the detection metrics but also increased computational complexity and the number of parameters. Among the evaluated ratios, $r_p = 1/2$ achieved the highest detection metrics, whereas $r_p = 1/8$ yielded the lowest computational cost. Compared with $r_p = 1/2$, $r_p = 1/4$ reduced GFLOPs from 11.55 to 11.07 and the parameter count from 4.46 M to 4.14 M, while P, R, mAP@0.5, and mAP@0.5:0.95 decreased by only 0.10, 0.07, 0.08, and 0.12 percentage points, respectively. Therefore, $r_p = 1/4$ was retained as the selected accuracy–efficiency trade-off for the subsequent experiments. Compared with Baseline, Model 1 with $r_p = 1/4$ reduced GFLOPs and the parameter count by 9.26% and 18.34%, respectively. Its effect on detection accuracy was mixed: mAP@0.5 increased by 0.25 percentage points, whereas P, R, and mAP@0.5:0.95 decreased by 0.23, 0.04, and 0.59 percentage points, respectively. Accordingly, C3k2_PartialNetBlock was interpreted primarily as an efficiency-oriented component that reduces computational redundancy while retaining broadly comparable detection performance, rather than as an independently accuracy-enhancing module.

As shown in Table 4, incorporating CIDAF into Model 1 increased P, R, mAP@0.5, and mAP@0.5:0.95 by 2.10, 0.77, 0.73, and 1.99 percentage points, respectively, indicating improved selective interaction and adaptive fusion between RGB appearance cues and pseudo-depth-derived structural information. The subsequent incorporation of BiFPN further increased P, mAP@0.5, and mAP@0.5:0.95 by 0.04, 0.17, and 0.46 percentage points, respectively, while reducing GFLOPs and parameter count by 13.24% and 16.89%. After SC-EUCB was introduced, the complete CIAFNet further improved the four detection metrics by 0.62, 0.18, 0.28, and 0.56 percentage points over Model 3, respectively, with only marginal increases in computational complexity. Compared with Baseline, CIAFNet improved P, R, mAP@0.5, and mAP@0.5:0.95 by 2.53, 0.83, 1.43, and 2.42 percentage points, respectively, while reducing GFLOPs and parameter count by 14.02% and 25.05%. Overall, C3k2_PartialNetBlock provides a lightweight feature extraction basis, while CIDAF, BiFPN, and SC-EUCB progressively improve cross-modal fusion, multiscale aggregation, and detail recovery, enabling CIAFNet to achieve a favorable accuracy–efficiency balance.

The paired bootstrap analysis in Table 5 further showed that CIAFNet improved mAP@0.5 and mAP@0.5:0.95 over Baseline by mean values of 1.34 and 2.64 percentage points, respectively. The corresponding 95% CIs were [0.71, 1.97] and [1.88, 3.40] percentage points. Because both confidence intervals were entirely above zero, the improvements over Baseline were statistically significant under the paired bootstrap criterion.

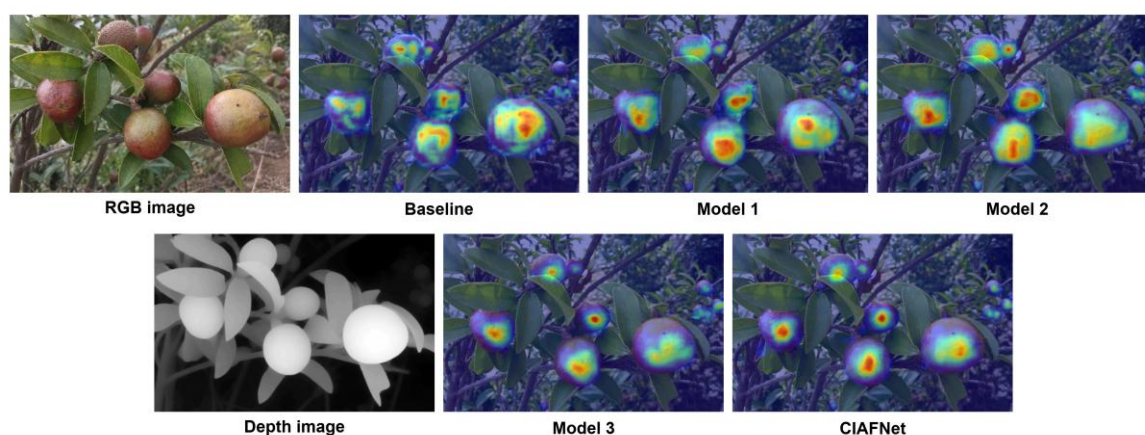
Table 5. Paired bootstrap comparison between Baseline and CIAFNet.

Metric	Baseline, Mean $\pm$ SD/%	CIAFNet, Mean $\pm$ SD/%	Mean Δ /pp	95% CI of Δ /pp
mAP@0.5	91.97 $\pm$ 0.45	93.31 $\pm$ 0.41	1.34	[0.71, 1.97]
mAP@0.5:0.95	71.14 $\pm$ 0.54	73.78 $\pm$ 0.50	2.64	[1.88, 3.40]

Note. CIAFNet denotes the model state obtained after first-stage training on the natural orchard RGB–pseudo-depth dataset. Δ denotes CIAFNet minus Baseline; pp denotes percentage points.

On the tested workstation, CIAFNet achieved a mean network-only inference latency of 10.716 ms per input, corresponding to approximately 93.32 FPS. In addition to theoretical complexity metrics such as GFLOPs and parameter count, this result provides an empirical reference for the forward pass efficiency of CIAFNet. It should be noted that the timing included only the network forward computation on the specified workstation and therefore does not directly represent end-to-end processing speed. Moreover, because embedded robotic platforms differ from the current workstation in computational resources, power constraints, and deployment environments, the actual inference performance on such platforms may vary and requires further evaluation through dedicated deployment tests.

Feature response heatmaps for the ablation models are shown in Figure 12. As C3k2_PartialNetBlock, CIDAF, BiFPN, and SC-EUCB were progressively introduced, high-response regions shifted from foliage and background areas towards the *Camellia oleifera* fruit bodies. Relative to Baseline, the enhanced models produced more concentrated responses around occlusion boundaries and adjacent fruit, clearer target boundaries, and weaker activation in non-target regions. This pattern is consistent with the quantitative improvements and qualitatively supports the capacity of the proposed modules to improve discriminative fruit region representations and spatial focus.

**Figure 12.** Heatmap visualizations for different module combinations in the ablation study.

3.4. Effectiveness of Pseudo-Depth Assistance

To examine the value of pseudo-depth as a structural prior and the effectiveness of CIDAF-based fusion, five configurations were compared: RGB only, pseudo-depth only, RGB plus pseudo-depth with feature concatenation, RGB plus pseudo-depth with feature addition, and CIAFNet with adaptive fusion.

To compare feature separability among fusion strategies, ROI features were extracted from intermediate feature maps after fusion and before the detection head and were visualized using t-SNE. The fused feature map with the highest spatial resolution was selected to balance post-fusion semantic representation with spatial localization information for small *Camellia oleifera* fruit targets. For each test image, bounding box regions

containing *Camellia oleifera* fruit and randomly sampled non-target background regions were mapped to the corresponding feature map. Average pooling within each ROI produced feature vectors for target and background regions. For each model, 260 fruit target ROI features and 295 background ROI features were extracted for t-SNE visualization and calculation of Sil and DBI. Detection performance is summarized in Table 6, and the corresponding target and background feature distributions are shown in Figure 13.

Table 6. Comparison of detection performance among pseudo-depth-assisted fusion methods.

Method	Input Modality	Fusion Strategy	P/%	R/%	mAP@0.5/%	mAP@0.5:0.95/%
RGB only	RGB	None	91.78	81.36	91.52	68.94
Depth only	Depth	None	84.86	61.20	71.24	53.91
RGB-D Concat	RGB + Depth	Feature concatenation	93.64	83.62	92.58	72.36
RGB-D Add	RGB + Depth	Feature addition	93.17	82.95	92.23	71.08
CIAFNet	RGB + Depth	CIDAF adaptive fusion	94.54	84.92	93.33	73.67

Note. In the table, CIAFNet denotes the model state obtained after first-stage training on the natural orchard RGB–pseudo-depth dataset.

Table 6 shows that the pseudo-depth-only configuration yielded P, R, mAP@0.5, and mAP@0.5:0.95 values of 84.86%, 61.20%, 71.24%, and 53.91%, respectively, all substantially lower than those obtained with the RGB-only configuration. Pseudo-depth therefore cannot independently replace appearance cues such as color, texture, and edges. When RGB and pseudo-depth were used jointly, RGB-D Concat and RGB-D Add increased mAP@0.5:0.95 to 72.36% and 71.08%, respectively, indicating that relative structural priors helped distinguish targets from foliage, background regions, and adjacent fruits. Compared with simple concatenation and element-wise addition, CIAFNet with CIDAF-based adaptive fusion achieved P, R, mAP@0.5, and mAP@0.5:0.95 values of 94.54%, 84.92%, 93.33%, and 73.67%, respectively. These values represented improvements of 2.76, 3.56, 1.81, and 4.73 percentage points over the RGB-only configuration.

The paired bootstrap results in Table 7 showed mean improvements of 1.86 percentage points in mAP@0.5 and 4.72 percentage points in mAP@0.5:0.95 relative to RGB-only. The corresponding 95% CIs were [1.21, 2.51] and [3.94, 5.50] percentage points, respectively. Neither confidence interval included zero, indicating that both improvements were statistically significant under the paired bootstrap criterion.

Table 7. Paired bootstrap comparison between RGB-only and CIAFNet.

Metric	RGB-Only, Mean $\pm$ SD/%	CIAFNet, Mean $\pm$ SD/%	Mean Δ /pp	95% CI of Δ /pp
mAP@0.5	91.45 $\pm$ 0.47	93.31 $\pm$ 0.41	1.86	[1.21, 2.51]
mAP@0.5:0.95	69.06 $\pm$ 0.57	73.78 $\pm$ 0.50	4.72	[3.94, 5.50]

Note. CIAFNet denotes the model state obtained after first-stage training on the natural orchard RGB–pseudo-depth dataset. Δ denotes CIAFNet minus RGB-only; pp denotes percentage points.

Figure 13 presents the t-SNE distributions of target and background features under the different fusion strategies, together with the Sil and DBI separability metrics. The RGB-only configuration yielded a Sil of 0.16 and a DBI of 2.23, whereas the pseudo-depth-only configuration yielded values of 0.09 and 3.02, respectively, indicating less stable inter-class boundaries when pseudo-depth was used alone. After pseudo-depth was introduced, Sil increased to 0.20 for RGB-D Concat and 0.18 for RGB-D Add, while DBI decreased to 1.96 and 1.99, respectively. CIAFNet achieved the highest Sil value of 0.24 and the lowest DBI value of 1.64, indicating more compact within-class representations and clearer between-class separation. These metrics provide feature space evidence supporting the selective-

fusion effect of CIDAF. t-SNE was used only to visualize the feature distributions and was not treated as independent validation of the internal mechanism.

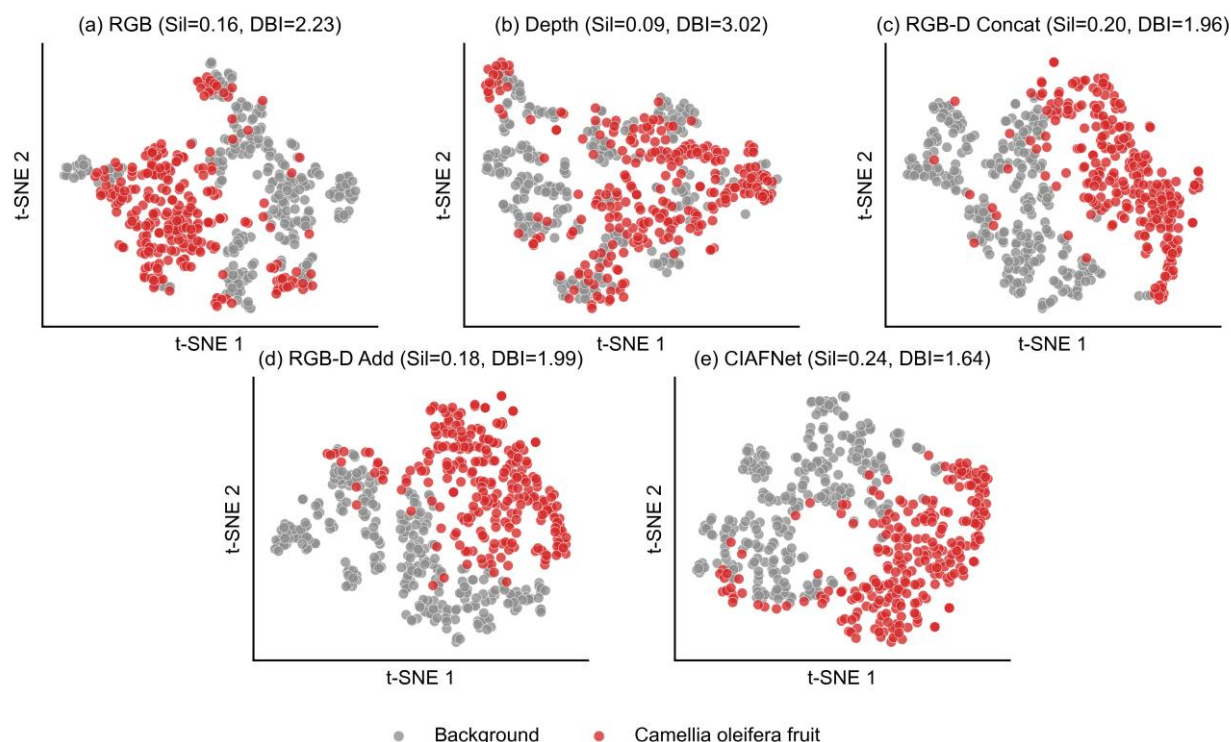


Figure 13. Feature-distribution visualizations for different pseudo-depth-assisted fusion methods.

3.5. Depth Input Discrepancy and Second-Stage Training Under Controlled Conditions

To quantify the effects of depth input source and second-stage adaptation under controlled conditions, four evaluation settings were constructed using the same D435i test set. First, the RGB-only baseline was evaluated using only the D435i RGB images. Second, CIAFNet (Stage 1), trained on the natural orchard RGB–pseudo-depth dataset, was evaluated using the D435i RGB images paired with DA-V2 pseudo-depth maps generated from the same images. Third, the same CIAFNet (Stage 1) model was evaluated by replacing the pseudo-depth maps with synchronized D435i-measured depth, without additional fine-tuning. Finally, CIAFNet (Stage 2), obtained by further fine-tuning CIAFNet (Stage 1) on the controlled scene D435i RGB-D dataset, was evaluated using D435i RGB images and measured depth. Because the second and third settings used identical model weights and RGB images, their comparison directly reflects the effect of replacing DA-V2 pseudo-depth with D435i-measured depth. Similarly, because the third and fourth settings used the same RGB and measured depth inputs, their comparison reflects the effect of second-stage measured depth adaptation. The results are presented in Table 8 and Figure 14.

Table 8. Detection performance with different depth inputs and second-stage training under controlled conditions.

Model Setting	Test Input Modality	P/%	R/%	mAP@0.5/%	mAP@0.5:0.95/%
RGB-only	RGB	95.12	89.36	95.31	77.68
CIAFNet (Stage 1)	RGB + Pseudo Depth	96.35	91.12	96.83	79.73
CIAFNet (Stage 1)	RGB + Measured Depth	94.33	88.21	94.17	75.92
CIAFNet (Stage 2)	RGB + Measured Depth	96.07	90.74	96.40	79.36

Note. CIAFNet (Stage 1) and CIAFNet (Stage 2) share the same network architecture and differ only in training state.

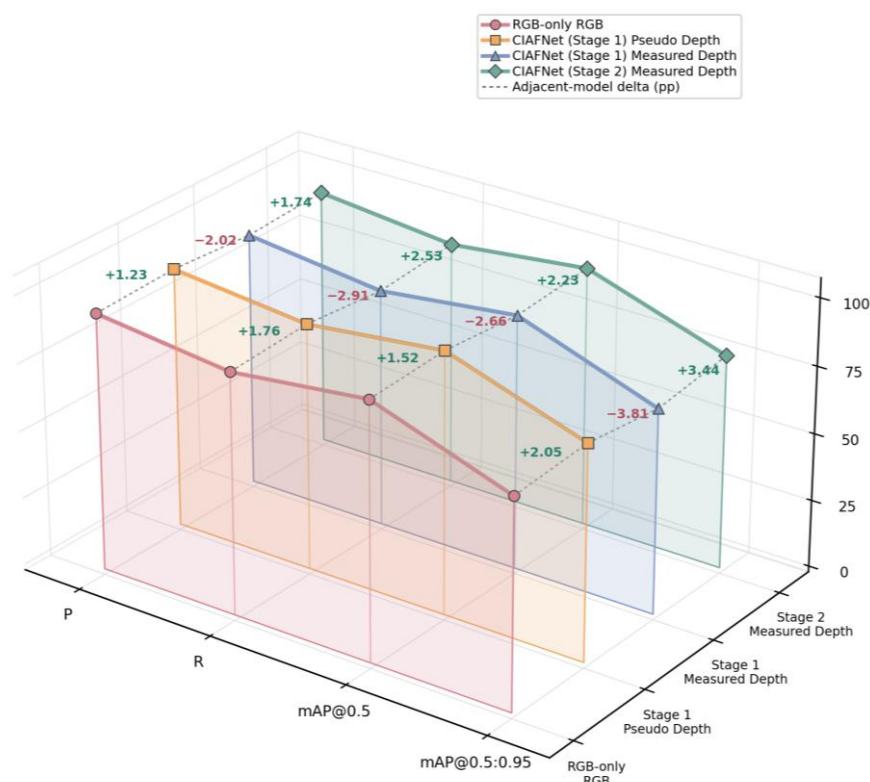


Figure 14. Comparison of detection performance with different depth inputs and second-stage training under controlled conditions.

When the D435i test set RGB images were paired with DA-V2 pseudo-depth maps, CIAFNet (Stage 1) achieved P, R, mAP@0.5, and mAP@0.5:0.95 values of 96.35%, 91.12%, 96.83%, and 79.73%, respectively (Table 8 and Figure 14). Compared with the RGB-only baseline, its mAP@0.5 and mAP@0.5:0.95 increased by 1.52 and 2.05 percentage points, respectively, indicating that the addition of pseudo-depth remained beneficial in the controlled scene test set. When the same CIAFNet (Stage 1) weights and RGB images were retained, but the DA-V2 pseudo-depth maps were replaced with synchronized D435i-measured depth, P, R, mAP@0.5, and mAP@0.5:0.95 decreased to 94.33%, 88.21%, 94.17%, and 75.92%, respectively. These values were 2.02, 2.91, 2.66, and 3.81 percentage points lower than those obtained with pseudo-depth input. Because the model weights and RGB images were unchanged, this degradation reflects the input domain discrepancy between DA-V2 pseudo-depth and D435i-measured depth, including differences in scale response, edge continuity, noise, and missing value patterns. After further fine-tuning on the controlled scene D435i RGB-D dataset, CIAFNet (Stage 2) achieved P, R, mAP@0.5, and mAP@0.5:0.95 values of 96.07%, 90.74%, 96.40%, and 79.36%, respectively. Compared with CIAFNet (Stage 1) evaluated directly with measured depth input, these values increased by 1.74, 2.53, 2.23, and 3.44 percentage points, respectively. This recovery indicates that second-stage fine-tuning adapted the feature representations and fusion weights to the statistical characteristics of D435i-measured depth, thereby mitigating the pseudo-depth-to-measured-depth input mismatch within the controlled data domain.

3.6. Relative Three-Dimensional Localization Consistency Under Controlled Conditions

Following validation of the detection performance, CIAFNet (Stage 2), obtained by fine-tuning on controlled scene D435i RGB-D data, was used to evaluate the relative displacement consistency of the 3D coordinates produced by the visual front end. Ten groups of relative displacement experiments were conducted under controlled conditions. In each group, five model *Camellia oleifera* fruit targets arranged on a simulated branch remained stationary, while the D435i camera was translated along the prescribed three-axis direction. The coordinates of the five targets were recorded before and after camera translation, yielding 50 target-level coordinate comparisons. The observed coordinate changes were compared with the expected changes induced by the prescribed camera translation. The experimental setup and an example of the coordinate output are shown in Figure 15, and the relative displacement consistency errors under controlled conditions are reported in Table 9.

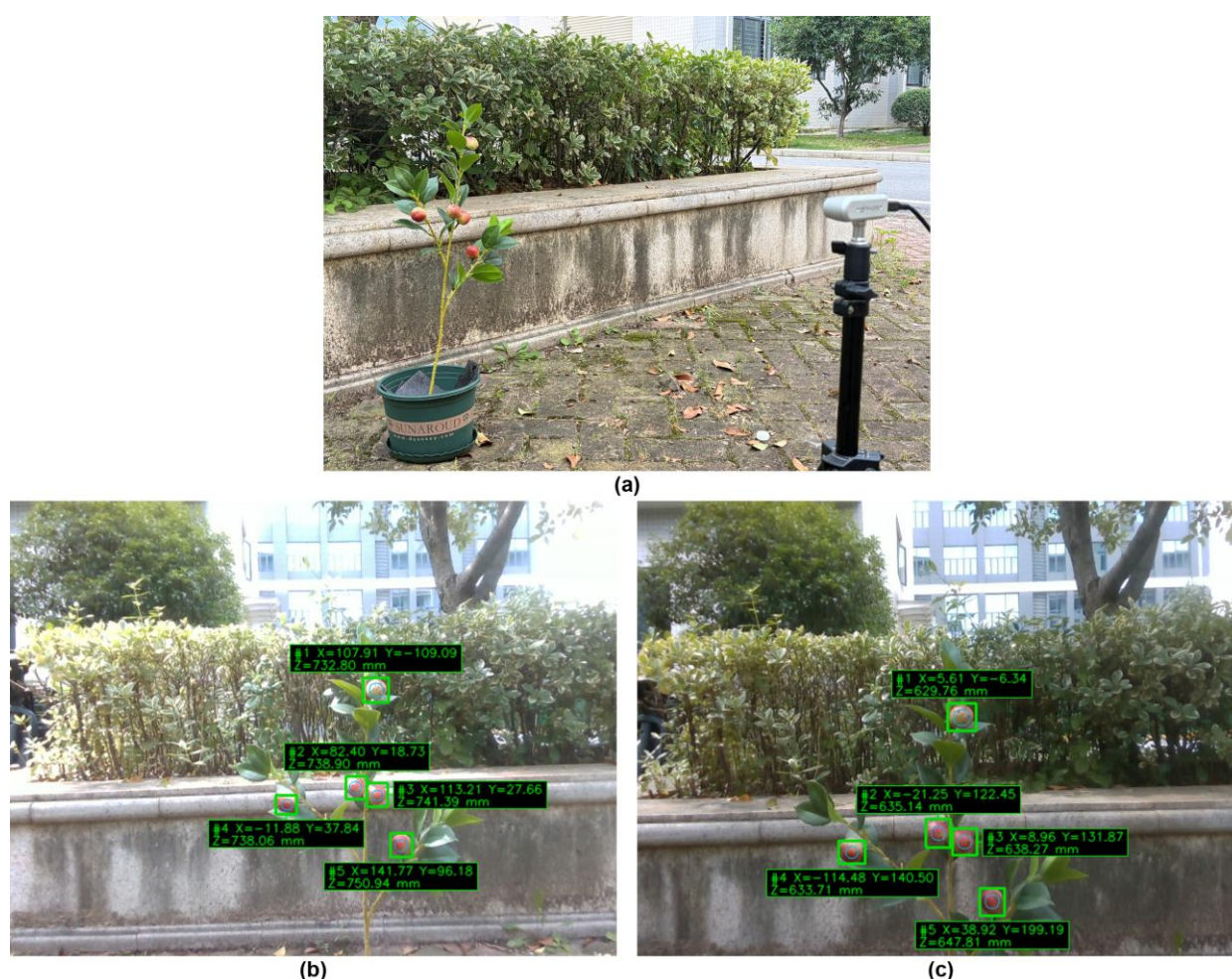


Figure 15. Experimental setup and 3D coordinate outputs before and after combined camera translation: (a) setup; (b) coordinates before translation; and (c) coordinates after simultaneous 100 mm rightward, forward, and upward translations.

Across the 10 experimental groups, comprising 50 target-level measurements, the pooled mean absolute errors along the X, Y, and Z axes were 3.10, 3.15, and 3.27 mm, respectively. The average of the three axis-wise mean absolute errors was 3.17 mm. These results indicate that, under controlled conditions involving model *Camellia oleifera* fruit targets and simulated branch structures, the CIAFNet-based visual front end produced relatively consistent coordinate changes under prescribed camera translations. Potential

sources of error included bias in the bounding box center, local depth holes within the ROI, depth discontinuities at target boundaries, and small pose perturbations during camera movement.

The mean 3D relative displacement consistency error across the 10 experimental groups was 5.60 mm, with the group-level mean values ranging from 4.94 to 6.20 mm, as shown in Figure 16. Although some variation among the tests was observed, the errors remained within a relatively narrow range. These findings support the feasibility and stability of using CIAFNet as a visual front end for relative 3D coordinate estimation under controlled conditions; however, they do not establish absolute localization accuracy in natural orchards.

Table 9. Relative displacement consistency results for *Camellia oleifera* fruit targets under controlled conditions.

Experimental Group	Number of Observations	MAE in X/mm	MAE in Y/mm	MAE in Z/mm	Mean 3D Consistency Error/mm
1	5	3.13	3.27	3.48	5.76
2	5	3.38	3.47	3.42	5.95
3	5	2.35	3.42	3.12	5.20
4	5	3.72	2.58	3.40	5.76
5	5	2.68	3.12	3.83	5.68
6	5	3.51	3.66	2.66	5.88
7	5	2.49	2.73	3.51	5.11
8	5	3.84	3.05	3.69	6.20
9	5	2.76	3.71	2.76	5.52
10	5	3.14	2.49	2.83	4.94
Overall	50	3.10	3.15	3.27	5.60

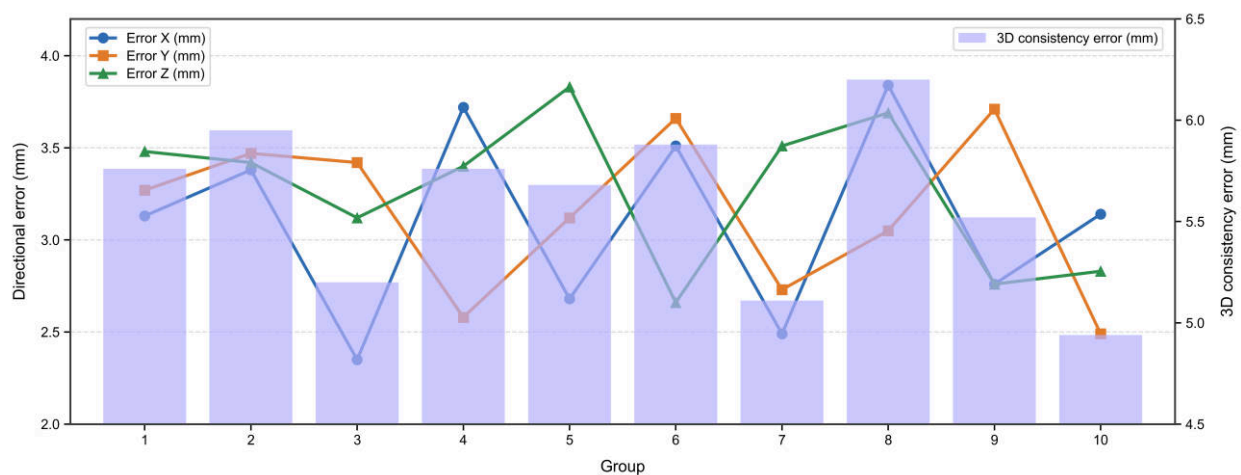


Figure 16. Statistics of relative 3D localization errors under controlled conditions.

4. Discussion

The central finding is that pseudo-depth improves *Camellia oleifera* fruit detection in natural orchards not by replacing RGB information but by providing relative structural cues that can be exploited selectively when appearance cues are ambiguous. CIAFNet embeds this prior within the detector rather than using it only after detection for coordinate calculation. Under the modality-matched RGB–pseudo-depth setting, CIAFNet (Stage 1) improved mAP@0.5:0.95 by 2.78 percentage points compared with YOLO-RGBDtea, the strongest multimodal baseline in Table 2. Separately, the within-framework comparison in Table 6 showed that CIAFNet (Stage 1) improved mAP@0.5:0.95 by 4.73 percentage

points over the RGB-only configuration. Pseudo-depth performed poorly when used alone but improved performance when combined with RGB; the gain is therefore more plausibly attributed to complementarity between appearance and structural cues than to replacement by either modality. Through discrepancy awareness, consistency modeling, and gated allocation, CIDAF enables selective use of depth structure around fruit boundaries, occlusion edges, and adjacent targets while suppressing unreliable responses caused by noise or holes. The ablation results, feature separability metrics, and visualizations collectively support this interpretation, although none independently proves every internal mechanism.

The experiments also revealed a substantial input mismatch between training with pseudo-depth and deployment with measured depth. DA-V2 pseudo-depth encodes relative structural relationships and can therefore support training on natural orchard images acquired without synchronized depth. Because these pseudo-depth maps are deterministically predicted from the corresponding RGB images rather than obtained through independent geometric measurements, their relative value distributions, boundary continuity, and noise characteristics may differ substantially from those of D435i-measured depth. The current results also cannot fully exclude the possibility that the depth branch partly relies on statistical patterns specific to DA-V2 outputs. Consistent with this interpretation, directly replacing DA-V2 pseudo-depth with D435i-measured depth reduced the detection performance of CIAFNet (Stage 1), indicating that the model is sensitive to the source and statistical characteristics of the depth input. After second-stage fine-tuning with synchronized D435i RGB-D data collected under controlled conditions, the performance with measured depth input improved substantially, indicating that CIAFNet (Stage 2) can adapt to D435i-measured depth without modification of the principal network architecture. However, the D435i dataset was collected using artificial fruit models and cut branches in a controlled scene and therefore does not encompass the full range of illumination, foliage occlusion, sensing distances, fruit surface properties, background structures, and depth quality conditions encountered in real orchards. For an active infrared sensor such as the D435i, the infrared reflectance of artificial fruit surfaces and leaves on cut branches may differ from that of living fruit and leaves. These material-dependent differences may affect the infrared pattern received by the sensor, thereby influencing missing depth patterns, target boundary quality, and depth stability. Accordingly, the second-stage results demonstrate adaptation to D435i-measured depth only within the current controlled data domain; they do not establish that CIAFNet (Stage 1) learned a representation invariant to the depth input source or that the measured depth performance will generalize to real orchards. These findings indicate that the source and statistical characteristics of the depth input should be treated as integral components of multimodal model design and deployment.

The localization experiment conducted under controlled conditions further suggests that stable detection regions can serve as a visual front end for relative 3D coordinate estimation. The controlled experiment comprised 10 groups and yielded 50 target-level relative displacement measurements. The mean 3D consistency error was 5.60 mm, with group-level mean values ranging from 4.94 to 6.20 mm. These results indicate only that the coordinate changes produced by the visual front end were relatively consistent under the prescribed camera translation within the current controlled scene. Nevertheless, all target measurements were obtained using artificial *Camellia oleifera* fruit models and cut branches under controlled conditions. Moreover, the evaluation compared the observed coordinate changes with those expected from the prescribed camera translation rather than comparing the estimated absolute target coordinates with independently measured ground truth positions. Therefore, the experiment evaluates relative displacement

consistency under controlled conditions and does not establish absolute positioning accuracy, localization performance on real fruits, or harvesting reliability in natural orchards.

The circular-ROI scaling coefficient and its adjustment range, as well as the learning rate, training budget, and parameter-freezing configuration used in the second stage, were empirically selected according to the requirements of the current task and the characteristics of the available datasets. Because exhaustive sensitivity analyses and systematic comparisons of alternative freezing strategies were not conducted, these settings should be regarded as task-specific design choices rather than universally optimal configurations.

In future work, we will conduct dedicated studies on the 3D localization of *Camellia oleifera* fruit by systematically evaluating the sensitivity of localization performance to the circular ROI scaling coefficient and investigating the stability of depth-based 3D coordinate estimation under different levels of target visibility and occlusion. Large-scale synchronized RGB-D data will also be collected from fruit on living *Camellia oleifera* trees in natural orchards, and absolute 3D positioning errors will be evaluated using an independent high-precision reference system under diverse illumination, occlusion, sensing distance, and depth quality conditions. End-to-end manipulator positioning and harvesting experiments will further be conducted to assess practical harvesting reliability.

5. Conclusions

This study proposed CIAFNet, a lightweight RGB-D dual-stream network for *Camellia oleifera* fruit detection in natural orchards. The principal methodological novelty of CIAFNet lies in its discrepancy- and consistency-aware cross-modal fusion mechanism. Unlike fixed feature concatenation or element-wise addition, this mechanism explicitly models local consistency and discrepancy between RGB and depth features to guide the selective fusion of RGB appearance information and DA-V2-derived relative structural priors. Specifically, C3k2_PartialNetBlock reduces redundant spatial computation through partial channel processing; CIDAF generates content-adaptive fusion weights to enhance complementary structural information and attenuate the influence of unreliable depth responses; and the SC-EUCB-enhanced BiFPN improves the recovery of fine-grained features across different scales. Together, these modules form a dual-stream detection framework that balances detection performance and computational complexity.

On the natural orchard RGB-pseudo-depth test set, CIAFNet (Stage 1) achieved a precision of 94.54%, a recall of 84.92%, an mAP@0.5 of 93.33%, and an mAP@0.5:0.95 of 73.67%, with 10.49 GFLOPs and 3.80 M parameters. The modality comparison and ablation experiments validated the effectiveness of pseudo-depth-assisted detection and the proposed network components. The image-level paired bootstrap analysis further showed that the performance improvements of CIAFNet over the RGB-only and Baseline configurations were statistically significant on the current test set. After second-stage fine-tuning using sensor-acquired D435i RGB-D data collected under controlled conditions, CIAFNet (Stage 2) showed improved adaptation to sensor-measured depth within the controlled data domain. In addition, the relative displacement consistency experiment conducted under controlled conditions yielded a mean 3D consistency error of 5.60 mm, indicating that CIAFNet detections could serve as a visual front end to support relative 3D coordinate estimation in controlled environments. However, the current conclusions apply only to pseudo-depth-assisted detection in natural orchards and to measured depth adaptation and relative localization under controlled conditions. Future work will use synchronized sensor-acquired RGB-D data collected in real orchards and continuous robotic harvesting experiments to further evaluate the end-to-end processing speed, robustness, absolute localization accuracy, and cross-environment generalization of CIAFNet.

Author Contributions: Conceptualization, Y.C. and H.A.; methodology, C.Y. (Chengxin Yang), Y.C. and H.A.; software, C.Y. (Chengxin Yang); validation, C.Y. (Chengxin Yang) and C.Y. (Chao Yuan); formal analysis, C.Y. (Chengxin Yang) and Y.C.; investigation, C.Y. (Chengxin Yang) and Y.F.; resources, H.A.; data curation, Y.L. and S.L.; writing—original draft preparation, C.Y. (Chengxin Yang); writing—review and editing, Y.C., C.Y. (Chao Yuan), Y.L., Y.F., S.L., H.A. and D.F.; visualization, C.Y. (Chengxin Yang); supervision, Y.C.; project administration, Y.C. and H.A.; funding acquisition, Y.C. and H.A. All authors have read and agreed to the published version of the manuscript.

Funding: This research was funded by the National Natural Science Foundation of China, grant number 51905387, and the Scientific Research Project of the Hubei Provincial Department of Education, grant number B2022316.

Institutional Review Board Statement: Not applicable.

Data Availability Statement: The datasets presented in this article are not readily available because they form part of an ongoing study. Requests to access the datasets should be directed to the corresponding authors.

Acknowledgments: During the preparation of this manuscript, the authors used ChatGPT (GPT-5.5; OpenAI, San Francisco, CA, USA) for Chinese–English translation, language polishing, and stylistic revision. The authors reviewed and edited all outputs and take full responsibility for the content of this publication. ChatGPT was not used for study design, data generation, data analysis, result interpretation, or figure creation.

Conflicts of Interest: The authors declare no conflicts of interest.

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